

Gait Module

User Manual

Module: Gait

Version: 1.5.0.158

Publication date: July 12, 2020

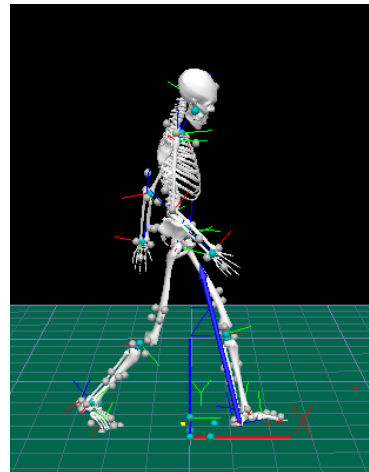
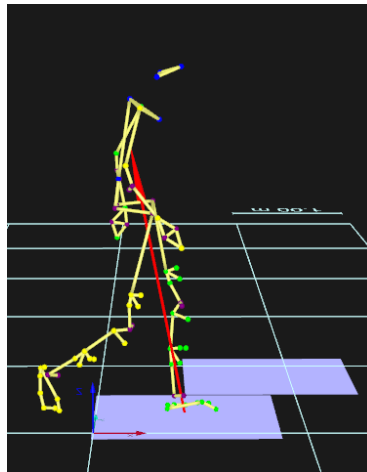


Table of Contents

Important information	6
Intended Use	6
Technical service and support	6
Scope of this manual	6
Introduction	7
Analysis steps	7
Installation	8
Software installation and updates	8
Required software versions	8
Installation of the processing engine	9
Folder setup	9
Visual3D	9
Theia3D	9
QCGM2	9
Open source CGM2	9
Motion Capture Cameras	10
Camera grouping	10
Video setup for web report	10
Offline report settings	10
Setting video file suffix	10
Setting up video time offsets	11
Online report	12
EMG setup	12
Capture and process a session	13
Step 1 – Getting Started	13
Step 2 – Computer Configuration (if required)	13
Step 3 – QTM Software	13
Step 4 – Camera Calibration	14
Step 5 – Subject and Session details	16

Step 6 - Prepare the subject	21
Step 7a – Collect data	22
Step 7b – Import existing files	24
Step 8 – Processing and preview report	25
Visual3D	25
QCGM2	26
Reports	27
Overview of report options	27
Visual3D Report	28
Reviewing data in Visual3D for quality assurance	28
Report graphs	29
Gait events	30
Modifying events in Visual3D	31
Modifying force assignments in Visual3D	32
Visual3D report customisations	33
Centre of Pressure plots (COP)	34
Word report	34
Graphs	34
Graph annotations	34
Word report customisations	35
Web report	35
Introduction	35
Playback	36
3D viewer	37
Video	38
Edit impairments	38
Details	39
Kinematic and kinetic timeseries	39
Temporal & spatial parameters	39
EMG data	39
Physical examination	40
Inserting physical examination data	40
Attachments	40
Attaching files to a report	41
Viewing attachments in a report	41

Gait features	42
Adding a feature without symbol	42
Adding a feature with symbol	43
Editing a feature	43
Reordering features	44
Impairments	44
Creating impairments	44
Editing impairments	45
Removing impairments	45
Findings	45
Workflow for adding features and impairments	46
Customising colours	46
Comparison report	47
Special analysis conditions	48
Inclined treadmill	48
Instrumented treadmill	48
Appendix A - EMG	50
Introduction	50
Naming convention for EMG channels	50
How to set up channel names	51
Noraxon	51
Delsys Trigno/Avanti	52
Myon / Cometa	52
Analog board	53
Appendix B - Normative data	54
Appendix C - Joint angles	58
Appendix D - Oxford Foot Model	60
Shank	60
Hindfoot	60
Reference	60
Forefoot	61
Reference	62
Hallux	62
Reference	62
Arch height	63

Reference	63
Foot maker placement compared to single segment foot	63
Single segment foot	63
Oxford Foot Model	64
Appendix E - Rizzoli Foot Model	66
Adding synchronized video to Visual3D	67
Revision history	70
Gait 1.5.0	70
Gait 1.4.3	71
Gait 1.4.2	72
Gait 1.4.1	72
Gait 1.4	73
Gait 1.3.2	74
Gait 1.3.1	75
Gait 1.3.0	75

Important information

Intended Use

This gait analysis module is intended for research use only and is not a Medical Device. For clinical use, Qualisys recommends the [Clinical Gait module](#). Clinical use is defined as performing measurements on patients for the purpose of diagnosis, treatment assessments or monitoring. If you would like to make a transition from a research gait module to the QCS, please contact support@qualisys.com.

Technical service and support

For information, visit our website:

www.qualisys.com

For technical service and support, contact us at:

Email: support@qualisys.se

Phone: +46 (0) 31 336 94 00

Scope of this manual

Qualisys marker-based technology is precise, robust and delivers high quality 3D-data indoor, outdoor or underwater (<http://www.qualisys.com>). Gait module also supports markerless technology using Qualisys video cameras and Theia3D software to obtain 3D-data.

For detailed information regarding camera setup, trajectory labeling (AIM) and basic operation of the system, please refer to the Qualisys Track Manager (QTM) manual. The PDF version of the QTM manual can be found in the Qualisys folder in the Windows Start Menu. Information about data processing with Visual3D can be found in the Visual3D Online Documentation, see Documentation section on www.c-motion.com. Information about data processing using Theia3D can be found in Theia3D online documentation, see Documentation section on www.theiamarkerless.ca.

Introduction

Analysis steps

Typically, the following steps are performed to carry out an analysis:

1. Switch on the system, wait for cameras to start up, prepare system for capture.
2. Start QTM and make sure the correct project is loaded (see top of QTM window)
3. Align L-frame with the force plates. Precise alignment is critical for the calculation of the joint moments/power. Either the X- or Y-axis must be parallel to the walking direction but it does not matter whether the subject walks in positive or negative X/Y-direction.
4. Calibrate.
5. If you use the C-Motion Caltester, perform a test to ensure the L-frame was aligned correctly and the force platform settings are correct.
6. Enter subject and session details in QTM.
7. Attach markers to subject.
8. Record static trial (all markers attached, ski pole position).
9. Remove 'static only' markers.
10. Record movement (dynamic) trials.
11. Label static trial and movement trials using Automatic Identification of Markers (AIM). Label markers manually and add trial to AIM model if not all markers were identified successfully.
12. Click Start Processing button to start the analysis.
13. Review results and print/export report.

Skip steps 7, 9 and 11 if using markerless approach.

Installation

Software installation and updates

A setup procedure must be performed before the module can be used or to perform updates. Instructions can be found in the QTM manual. (Start QTM, press F1 and enter PAF in the search box.)

Note: in Project options > Processing > C3D export, the option "Use zero force baseline correction" must be unchecked to make sure that force data can be processed correctly.

Required software versions

The current version of the module has been developed and tested for:

- Analysis module: 1.5.0.158
- QTM: 2020.1 build 5520
- Visual3D: 2020.05.1
- Theia3D: 2020.03.0.955
- QCGM2: 1.0.0 (based on pyCGM2 3.2.15)
- Report Generator: 0.8.5710.13161
- Microsoft Office 2016 (64-bit)
- Google Chrome Browser 83.0.4103.97 (Official Build) (64-bit)



Using the module with other versions may result in errors.

To check that the correct software versions are installed:

1. QTM: select Help > About. If you need to install a new version of QTM, log on to the [Qualisys website](#) with your user name and license key (see Help > About ...) to download the correct version and install it.
2. Visual3D: select Help > About Visual3D. If the computer is connected to the internet, select Help > Check for Updates. If an update is available, log on to the [C-Motion website](#) with your user name and password (supplied from C-Motion) to download the new version and install it.

3. Theia3D: select View > Info. If you need to install a new version of Theia3D, go to [Theia Markerless website](#).
4. QCGM2: Installer always contains up-to-date version of QCGM2. When analysis module is installed, latest version of QCGM2.exe is copied to C:\Program Files\Qualisys\QCGM2\QCGM2.exe. To check what version you use, right-click on executable, select Properties, select Details sheet and check Product version.

Further installation instructions can be found in the QTM manual. (Start QTM, press F1 and enter PAF in the search box.)

Installation of the processing engine

The biomechanics calculations are performed by Visual3D. Visual3D is not part of the installer and must be installed separately.

Theia3D must be also installed separately, if you want to use markerless setup. Note that Visual3D is still required when using Theia3D to obtain raw kinematics data..

CGM2 session type is processed by QCGM2 engine. For details about CGM2, please refer to <https://pycgm2.github.io/>.

Folder setup

Visual3D

1. Follow the steps from the C-Motion Wiki to install and activate Visual3D.
2. In QTM, go to Project Options > Miscellaneous > Folder Options and set the path for Visual3D to C:\Program Files\Visual3D\Visual3D.exe.

Theia3D

1. Follow the steps from the Theia Markerless Documentation to install and activate Theia3D.
2. In QTM, go to Project Options > Miscellaneous > Folder Options and set the path for Theia3D to C:\Program Files\Theia\Theia3D\Theia3D.exe.

QCGM2

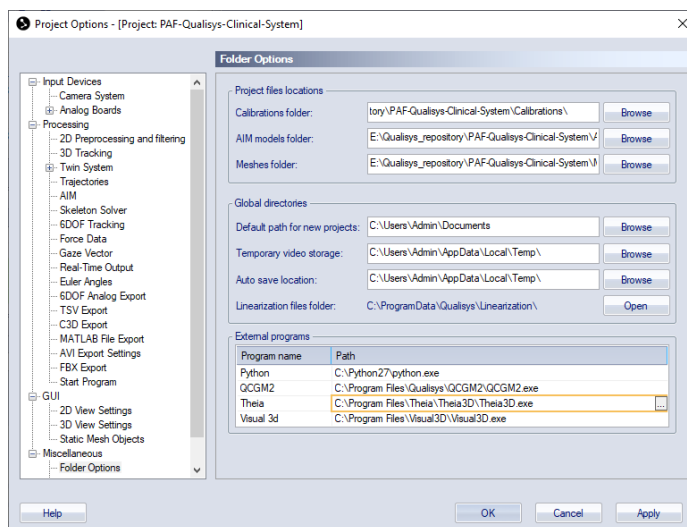
1. In QTM, go to Project Options > Miscellaneous > Folder Options and set the path for QCGM2 to C:\Program Files\Qualisys\QCGM2\QCGM2.exe.

Note: QCGM2 is based on an open source project pyCGM2 (see <https://pycgm2.github.io/>). For the convenience we use compiled version of the project in PAF module.

Open source CGM2

It is also possible to run the Python code directly with analysis module. This could be useful for customers who want to modify QCGM2 functionality. Please refer to <https://pycgm2.github.io/pages/qtmInstallAddon.html> for installation instructions.

Optionally you can use open source project from Qualisys GitHub (https://github.com/qualisys/qualisys_CGM2_workflow).



Motion Capture Cameras

The recommended measurement frequency (frame rate) for the motion capture cameras (both marker-based and markerless) is 100 Hz.

Camera grouping

In the situation when flash of one camera reflects from a shiny surface and appears on the other camera and other means on removing these reflexions fail, the exposure delay setting can be applied to shift the exposure of a camera compared to other cameras in the system. However, because the time of the exposure will then be different in the camera system it may be harder to get a good tracking.

To activate the exposure delay, select Camera group or Advanced on the Exposure delay mode setting.

Exposure delay	
Exposure delay mode*	Camera group
Camera groups	Camera group 1

For more details on camera grouping, please refer to QTM manual (in QTM press F1 and search for "exposure delay").

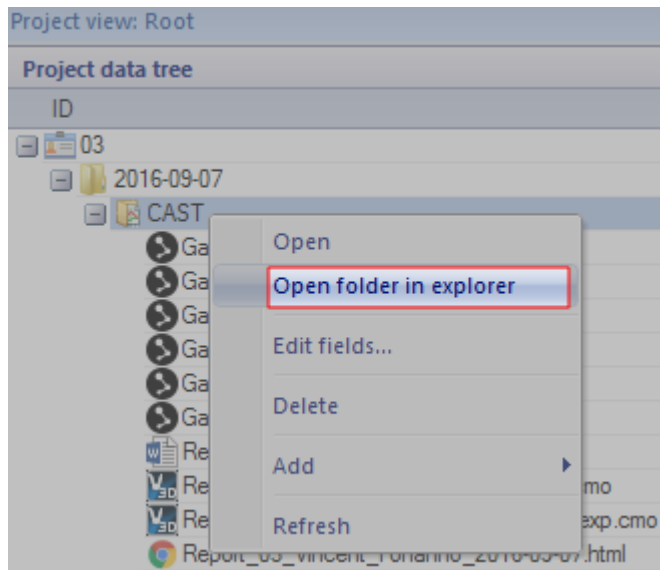
Video setup for web report

Offline report settings

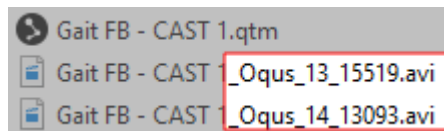
Setting video file suffix

If you are using video cameras, you should set the correct suffix for the frontal and sagittal video cameras:

1. After setting up the cameras, capture a test file. Save the file.
2. Open the folder containing the file by right-clicking it in the Project data tree:



3. Take note of the video file name suffix for the .avi files. Play the videos if you are unsure which video is frontal and sagittal:



4. Open `[Project folder]\Settings\User Settings.paf` in a text editor.

Edit the suffix settings to match your video file suffixes:

```
Frontal video: '_Oqus_13_15519.avi'
Sagittal video: '_Oqus_14_13093.avi'
```

Note that if you set the suffixes to `default`, the first and second video file found will be used as the sagittal and frontal video respectively:

```
Frontal video: 'default'
Sagittal video: 'default'
```

Setting up video time offsets

If your video camera is not frame synchronized, it can help to set a time offset for your camera. This is done by right clicking your live video in QTM.

At this time, this module cannot read the time offset you set in QTM, therefore, to get properly offset video in the web report you must also set the time offset in the *User Settings.paf* file.

The workflow is as follows:

1. Figure out the best time offset for your camera. Set it in QTM:
2D > Right click video > Set time offset.
2. Open `[Project folder]\Settings\User Settings.paf` in a text editor.

Edit the time offset settings with the same values set in step 1. Example: if your optimal time offset is 0.3 seconds for both cameras, set them both to 0.3.

```
Sagittal video offset: 0.3
Frontal video offset: 0.3
```

Online report

Similarly to offline report, videos can be adjusted in online report as well. Open [Project folder]\Settings\User Settings.paf in a text editor.

1. Set output video quality to values between 1 and 100 (the higher the value, the better quality and larger file size):

```
Video quality: 50
```

2. If camera view is rotated, rotation can be adjusted by modifying following settings:

```
Cameras:
  Include only defined cameras: True
  12345:
    Rotation: 90
    Group: Sagittal
  54321:
    Rotation: 270
    Group: Frontal
```

Number is serial number of camera the settings is being applied to. Camera serial number is part of .avi file name (see above in offline settings). The rotation applied to camera is set in degrees clockwise in 90 degree increments. Note that after modifying the video rotation, it is necessary to delete the mp4 files from session folder before generating the online report again.

Group is an optional setting. All cameras with the same group will be grouped together.

By default all videos will be shown in the report. If Include only defined cameras is set to True, only camera whose serial number is specified will be shown in the report.

EMG setup

Set up the channel names according to “Appendix A - EMG” on page 50.

Capture and process a session

Step 1 – Getting Started

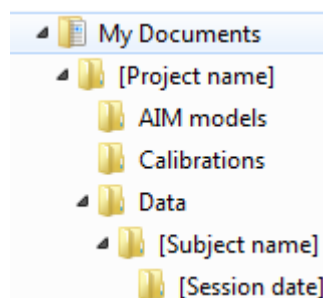
1. Turn the cameras on. The cameras will need a few seconds to start up. You will know when the cameras are ready when you can see numbers or the Qualisys logo underneath the lens or indicator light is green (depending on type of camera).

Step 2 – Computer Configuration (if required)

1. Right click on the Qualisys DHCP server in the task bar at the bottom right of the desktop.
2. Choose Qualisys configuration wizard and double-check that the network interface is configured for Oqus.
3. In QTM, choose Project Options > Input devices > Camera System > Connection and locate the camera system.

Step 3 – QTM Software

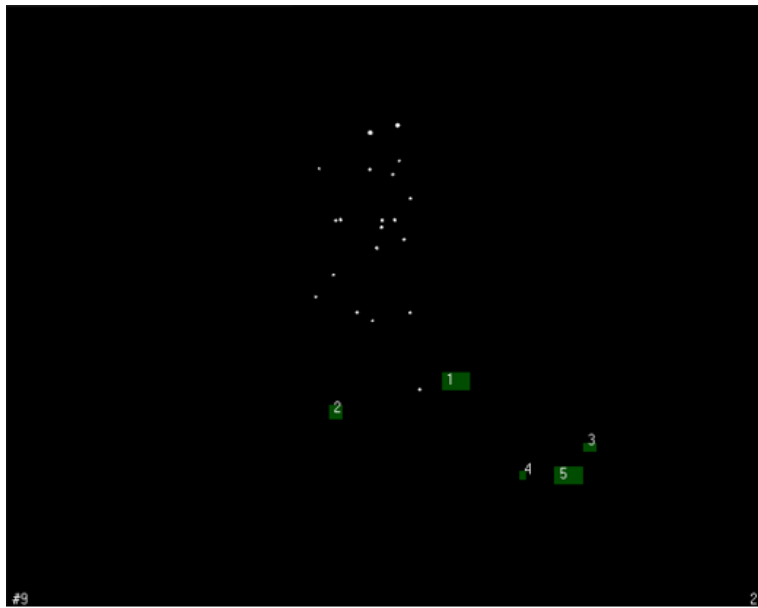
1. Start the QTM software, for example from the shortcut on the desktop.
2. Select the correct project, for example Gait module.
3. If the Project view is not shown on the left-hand side of the screen, select View > Project view or press Ctrl + R.
4. All trials are collected via the Project view:
 - a. All trials and sessions are added to one Project (for example “Gait module”) – it is not recommended to create a new project for each subject.
 - b. Trials have to be collected via the red/green buttons (see below) in the Project view and can be found on the hard drive in this structure:



Step 4 – Camera Calibration



1. Click on the New Page icon
2. Enter 2D view and first physically remove all reflective objects, such as the calibration kit or similar reflective equipment, from the measurement volume.
 - a. Go to the Camera settings sidebar on the right side of the 2D view. Click on Enable for Marker masks.
 - b. Click on “Auto-create” and start. QTM will mask all unwanted reflections. For manual masking options and advanced handling of camera settings and marker masks, please refer to the QTM Manual.



3. Place the L-shaped calibration reference on the floor.



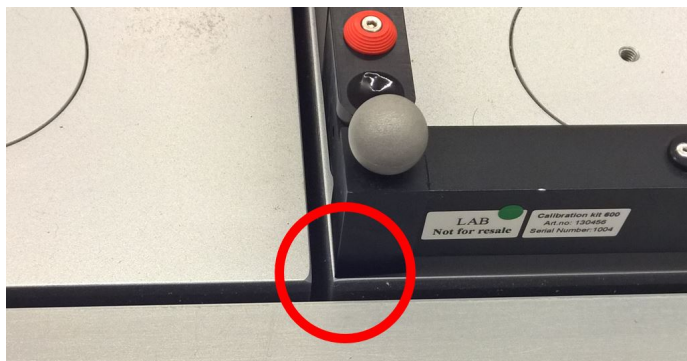
Precise alignment is critical for the calculation of the joint moments/power.

The correct placement depends on the convention that was used when setting up the force plates in QTM. Either the X- or Y-axis must be parallel to the intended walking direction but it does not matter whether the subject walks in positive or negative X/Y-direction.

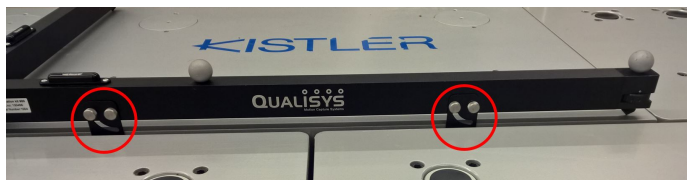
- a. The L-frame must be always placed on the same force plate, and in the same corner. We recommend that you take a picture of the default placement of the L-frame and put that picture in your lab.



- b. The corner of the L-frame must be placed exactly on the corner of the force plate.



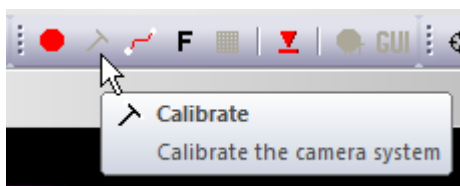
- c. The long axis of the L-frame must be aligned exactly with the edge of the forceplate. If your L-frame has force plate alignment aids, use them for better alignment. The alignment of the short axis is not critical.



- d. Always use height adjustment screws so that both axes are horizontal (use the spirit level gages for guidance). Avoid changing the setting of the screws between calibrations.



4. To calibrate, click on the calibration tool on the top menu bar.



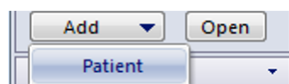
5. A window will open that will ask how long you would like to capture. 20-30 seconds is a sufficient amount of time for a standard full body volume. For large volumes, increase the time so that you have enough time to cover the complete volume that will be used.
6. Click start and you will hear a bell. Using your calibration wand walk around the capture area and wave the wand in the area that the subject will be moving in. Note: for guidance how to move wand in the volume refer to QTM user manual in the section Making a measurement > Calibration of the camera system > Wand calibration method.
7. Stop when you hear another bell.
8. Confirm that the average camera residual is approximately 1-2 mm or under. Each camera should have collected at least 300 points. The standard deviation of the wand length should be 1-2 mm or better for a full body volume.
9. If you use the C-Motion Caltester, now you can perform a test to ensure the L-frame was aligned correctly and the force platform settings are correct.

Step 5 – Subject and Session details

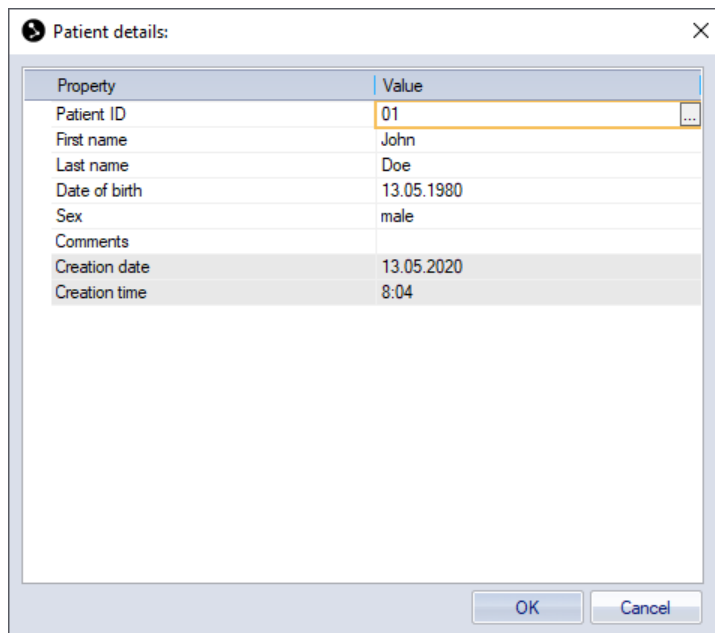


Make sure the correct patient's details are used.

1. Add a new patient or select an existing patient from the data tree.



2. Enter patient details:

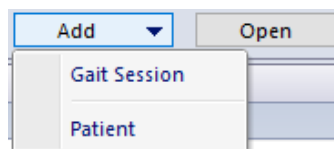


A dialog box titled "Patient details:" with a close button (X) in the top right corner. It contains a table with two columns: "Property" and "Value". The table has the following data:

Property	Value
Patient ID	01
First name	John
Last name	Doe
Date of birth	13.05.1980
Sex	male
Comments	
Creation date	13.05.2020
Creation time	8:04

At the bottom of the dialog box are two buttons: "OK" and "Cancel".

3. Click Add and add a Gait Session:



A close-up of a button labeled "Add" with a downward arrow. A dropdown menu is open, showing two options: "Gait Session" and "Patient".

4. Update the session details for the patient.

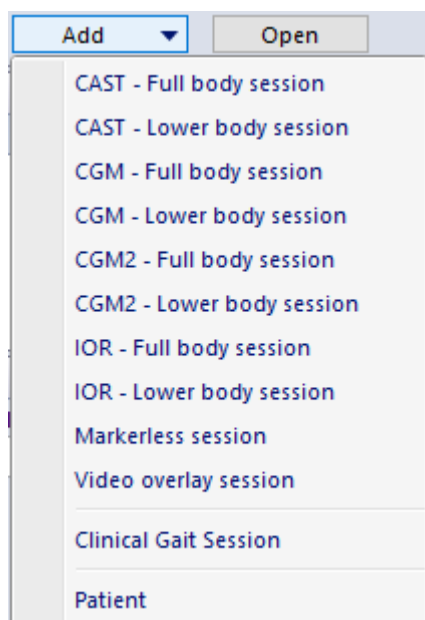
Property	Value
Patient ID	01
First name	John
Last name	Doe
Case ID	
Test operator	unspecified
Comments	
Session date	13.05.2020
Creation date	13.05.2020
Creation time	8:47
Diagnosis	unspecified
Secondary diagnosis	unspecified
Affected side	unspecified
Gross Motor Function Classification	unspecified
Functional Mobility Scale	unspecified
Height	0.00 m
Weight	0.0 kg
Leg length left	0 mm
Leg length right	0 mm
Knee width left	0 mm
Knee width right	0 mm
Ankle width left	0 mm
Ankle width right	0 mm
Sole delta left	0 mm
Sole delta right	0 mm
Shoulder offset left	0 mm
Shoulder offset right	0 mm
Elbow width left	0 mm
Elbow width right	0 mm
Wrist width left	0 mm
Wrist width right	0 mm
Hand thickness left	0 mm
Hand thickness right	0 mm
Marker diameter	14.0 mm

The following fields are relevant for the data processing:

- a. Weight: body weight is used to normalize joint moment and power curves.
- b. Ankle width left/right (required for legacy Conventional Gait Model (CGM) and for new CGM2 (version 1.0 and 1.1)): medio-lateral width of the ankle at the malleoli; used in the definition of the ankle joint centre. Note: CGM2 model version 2.1 and later do not require this metric as medial marker is used instead.
- c. Knee width left/right (required for legacy Conventional Gait Model (CGM) and for new CGM2 (version 1.0 and 1.1)): medio-lateral width of the knee at the epicondyles; used in the definition of the knee joint centre. Note: CGM2 model version 2.1 and later do not require this metric as medial marker is used instead.
- d. Leg length: (required for CGM2, optional for other marker sets) It is used for the normalisation of tempo-spatial data. If value is set to zero (default) length is taken from the marker distance from the ASIS to medial malleolus. For patients that are unable to stand with the knees fully extended in the static trial (e.g. fixed flexion contractures), the value needs to be measured and entered manually.

5. Create a sub-session: each condition that you are testing within a session requires a separate sub-sessions (e.g. barefoot, with shoes, etc.). When just testing a single condition, simply create a single sub-session only.

When creating the sub-session, you decide which marker set will be used and whether it is a full-body or lower-body session:



Full body session includes the thorax and the arms, lower body session includes only the pelvis and the legs.

Markerless algorithm always tracks the whole body.

6. Enter the sub-session details:

Property	Value
Patient ID	01
Case ID	
First name	John
Last name	Doe
Test condition	unspecified
Comments	
Creation date	13.05.2020
Creation time	8:50
Prothesis_Orthosis	No help
External aid	No help
External aid side	unspecified
Personal aid	No help
Personal aid side	unspecified
Event mode	Multiple forceplates
Left foot normalised to static trial	☒
Right foot normalised to static trial	☒
Multisegment foot	None
Normative data	Default
Update existing web report	☒

The following fields are relevant for the data processing:

Note that not all fields are available for all sub-session types.

- a. Test condition: used to identify different conditions when comparing sub-sessions. Enter a word or phrase that describes the test condition (barefoot, with shoes, with orthosis, ...).
- b. Ankle width left/right: see session details above (relevant for Conventional Gait Model only).
- c. Knee width left/right: see session details above (relevant for Conventional Gait Model only).
- d. Leg length left/right: see session details above.
- e. Event mode (not available for CGM2):
 - i. Manual: events must be identified by hand in QTM, useful if no force plates are present or if pathological gait makes it impossible to use automatic event recognition in Visual3D.
 - ii. Multiple force plates (default): Visual3D will use automatic event recognition for both legs). By default, the reports show all gait cycles in consistency graph. For Word report there is an option to delete all events prior to the first force plate contact and after one stride for each body side. To turn this on, open the file [project folder]\Templates\settings.php and set the setting for '\$auto_range' to 'true'.
 - iii. Single forceplate: Visual3D will prompt the user to select one template file for the right and left leg. In the template file, the corresponding foot must make contact with the force plate.
 - iv. No forceplate (using template file): will prompt the user for one template file for the right and left leg. In the template file, you must manually add a single left heel strike (LHS), left toe off (LTO), right heel strike (RHS) and right toe off (RTO) event before executing the analysis. It is possible to use the same template file for left and right or to use two different files.
 - v. No forceplate (automatic): uses relations between pelvis and heel to detect heel strikes and toe off. Note: this event mode may not work properly with pathological gait. For pathological gait, "No forceplate (using template file)" should be preferred instead. The algorithm that is used is described in this paper (Coordinate Based Algorithm): Zeni JA, Richards JG, Higginson JS (2008) Two simple methods for determining gait events during treadmill and overground walking using kinematic data. *Gait & Posture* 27, 710-714
 - vi. Instrumented treadmill (split belt): will detect events based on the force data for all steps. Requires (1) that the treadmill speed is entered by the user for each recorded file and (2) that the feet are on separate belts of the treadmill at all times. If both feet are on the same belt, the force assignment to the feet will be incorrect and events cannot be detected. Please navigate to [project folder]\Templates\settings.php and set the setting for \$treadmill_type = 'AMTI_tandem'; Currently only AMTI Tandem treadmill is supported. Please contact support@qualisys.com if you need other treadmill to be implemented.
 - vii. Instrumented treadmill (single belt): events are detected based on the acceleration of the COP. Requires that treadmill speed is entered by the user for each dynamic trial. Inverse dynamics results are only valid while a single foot is in contact with the treadmill. Please navigate to [project folder]\Templates\settings.php and set the setting for \$treadmill_type = 'Treadmetrix'; Currently only Treadmetrix treadmill is supported. Please contact support@qualisys.com if you need other treadmill to be implemented.
- f. Left/right foot normalised to static trial: if the box is checked, the foot is assumed to be flat on the floor in the static trial. The long axis of the foot will be assumed to be parallel to the floor. The height of the heel and toe markers will not affect the slope of the long axis of the foot.
- g. Multisegment foot (not available for CGM2): Two multisegment foot models are supported: Oxford Foot and Rizzoli Foot. If set to one of these options, the multisegment foot model will

be appended to the model and related graphs added to the reports. Further information can be found in “Appendix D - Oxford Foot Model ” on page 60 or “Appendix E - Rizzoli Foot Model ” on page 66.

- h. Normative data: you can define specific sets if normative data for different age groups. Further information on how to generate normative data sets can be found in “Appendix B - Normative data” on page 54. File names for user-generated normative data files can be adapted in Templates/settings.php.

Note: Normatives cannot be selected for CGM2. Normatives shown for CGM2 model are based on paper published by Schwartz in 2008, "The effect of walking speed on the gait of typically developing children". Data are available at Excel format at <https://github.com/pyCGM2/pyCGM2/tree/Master/pyCGM2/Data/normativeData>. Subset with 'Spontaneous speed' is used.

- i. CGM2 model: (Available for CGM2 only.) This field allows to change applied CGM model. It also includes option to choose original CGM model (CGM1.0). Please refer to <https://pycgm2.github.io/> for details about various versions of the CGM model. By default the CGM2.3 model is applied.

CGM2 Model	CGM2.3-skinClusters
Patient ID	CGM1.0
First name	CGM1.1
Last name	CGM2.1-HJC
Height	CGM2.2-IK
Weight	CGM2.3-skinClusters
	CGM2.4-ForeFoot

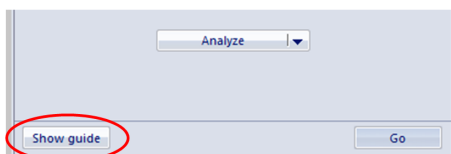
- j. Update existing web report: By default each time the report it uploaded to report center, previous report is replaced by latest upload. If checkbox is deselected, each upload is saved as separate report.

Step 6 - Prepare the subject

Apply the markers to the subject. Note: for quality and consistency of data it is important to pay close attention to marker placement and apply them at correct anatomical positions.

Always prepare a sufficient number of markers before the person you are testing arrives. If markers do not stick well, use a razor to remove some hair before attaching them. You can also contact Qualisys for stronger adhesive.

To show a document with the marker locations, click on Show Guide at the bottom of the Project Automation pane after you created the session:



Alternatively, you can find the marker guides in PDF format in the following location: [Project folder] \Documentation\...\pdf.

Note that head (SGL, L_HEAD, R_HEAD) and hand (L_HM2, R_HM2) are optional for the CAST and IOR models:

- If the markers SGL, L_HEAD and R_HEAD are present in the static and dynamic files, a head segment will be included in the body model.

- If the markers L_HM2 and R_HM2 are present in the static and dynamic files, hand segments will be included in the body model.

Step 7a – Collect data

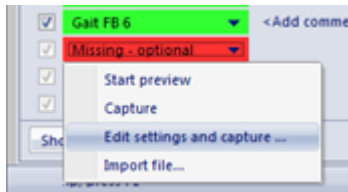
Note: if the options to create a new patient/session are not shown, select View > Project View.

Proceed to step 7b if you want to import existing files.

1. Begin the capture by having the subject stand in the middle of the field of view.
 - Elbows bent approximately 90° as if holding ski poles or riding a motorcycle.
 - Legs approximately shoulder-width apart.
 - Feet flat on the floor. If a patient cannot place their feet flat on the floor, be sure to uncheck the option "Left/right foot normalised to static trial" (see "Step 5 – Subject and Session details" on page 16)

For marker-based sessions, there are two types of static trial available: Static Anterior and Static Posterior. Only one of them is required. If both static trials exist and are selected, anterior one is used.

2. Click on the red box for the static file and select Capture to begin capturing.



Note: if you want to change the duration of the recording, select Edit settings and capture instead. QTM will remember the time you enter and will use it for all future trials you are recording for the measurement type within this project.

3. Review the capture to ensure all markers are present. The subject must be completely static. If required, crop the file to a section in which the subject was static and all markers were visible.
4. Remove 'static only' markers.
5. Repeat steps 2. and 3. to record gait trials. When using force plates, files should start approximately one step before the first force plates and end 2.5-3 steps after the last force plate.
6. Should the AIM fail to identify all markers, label manually and add the file to the AIM model. See the QTM manual for details. It is recommended that you have one 'master' AIM model for all subjects and keep adding trials from different subjects to that model if the model does not identify all markers correctly. The following AIMs are used:
 - a. CAST:
 - i. Lower body only:
 - i. Static: CAST_Static_Lower_Body.qam
 - ii. Gait: CAST_Dynamic_Lower_Body.qam
 - ii. Full body:
 - i. Static: CAST_Static_Full_Body.qam
 - ii. Gait: CAST_Dynamic_Full_Body.qam

- b. Leardini / IOR:
 - i. Lower body only:
 - i. Static: IOR_Static_Lower_Body.qam
 - ii. Gait: IOR_Dynamic_Lower_Body.qam
 - ii. Full body:
 - i. Static: IOR_Static_Full_Body.qam
 - ii. Gait: IOR_Dynamic_Full_Body.qam
- c. Conventional Gait Model (CGM):
 - i. Lower body only (static and dynamic): CGM_Lower_Body.qam
 - ii. Full body (static and dynamic): CGM_Full_Body.qam
- d. Oxford Foot Model: same as above but with suffix "_OxfordFM"
- e. Rizzoli Foot Model: same as above but with suffix "_RizzoliFM"
- f. Conventional Gait Model 2.i (CGM2):
 - i. Full body:
 - i. Static: CGM2_Static_Full_Body.qam
 - ii. Gait: CGM2_Dynamic_Full_Body.qam

The names of the AIM models must match exactly so that QTM will select them automatically when you capture a file. Overwrite these models with new versions if required.

- 7. Check that all markers are present and labeled correctly.
- 8. When using the CGM2 session type, mark any invalid force plate contacts as follows:
 - a. Identify the force plate number with the invalid contact. The force plate name is shown when you hover over the force plate with the mouse in the QTM 3D view. If you are unsure about the order of force plates, review the force plate names in QTM > Project Options > Force data.
 - b. Right-click measurement in the Project data tree and select "Edit fields".

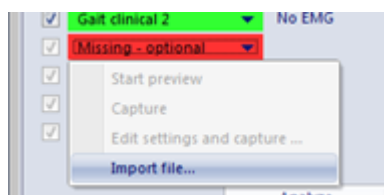
- c. Set the force plates that you want to modify to "invalid", "left" or "right":

Details for item 'Gait - CGM2 1.qtm':	
Property	Value
Forceplate1	Auto
Forceplate2	Auto
Forceplate3	Auto
Forceplate4	Auto
Forceplate5	Auto
Comments	
Patient ID	003a
First name	John
Last name	Doe
Creation date	26.05.2020
Creation time	12:37
Case ID	
Marker lowpass filter frequency	6.0 Hz
Marker lowpass filter order	4
Forceplate lowpass filter frequency	0.0 Hz
Forceplate lowpass filter order	0

9. Lowpass filter can be applied to marker data and to analog data. For CAST, IOR and CGM and Markerless session, the filter is applied globally (for all patients). To change the filter, navigate to [your project]\Templates\settings.php and adjust parameters at Filtering section. For CGM2 session type, lowpass filter is set individually for each dynamic measurement. Modify it by right-clicking the measurement in the Project data tree and select "Edit fields" and specify filtering parameters (see picture above).
10. If walking on treadmill, enter treadmill speed at measurements details. If speed is set to 0 (default), treadmill speed will be calculated from minimum foot velocity during contact.
11. Proceed with Step 8.

Step 7b – Import existing files

1. If you want to process existing QTM files (that you recorded outside analysis module), click on the red box for the corresponding file and select Import file.



The files will be copied into the session folder and renamed to a standard name. Use the comments field if you want to make a note of the original file name.

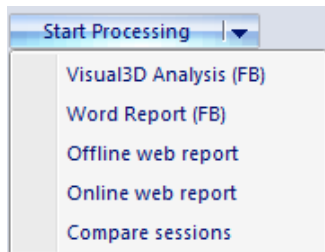
2. Repeat for all required files. Check if files are labeled correctly and crop if needed. If you need to label the files with AIM, see Step 7a and the QTM User Manual for details. Please note that video files are not imported automatically and must be copied manually to working folder.

Step 8 – Processing and preview report

Visual3D

When the minimum required number of files has been captured, the Start processing button is available. Click the small triangle to see the different analysis options:

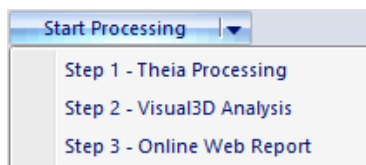
Marker-based session:



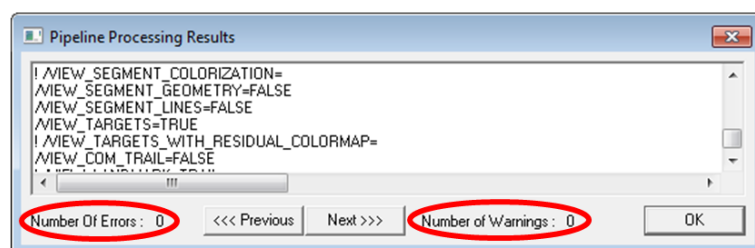
Start the analysis by selecting the top option (Visual3D Analysis). When Visual3D analysis is clicked, QTM automatically exports any file that has its checkbox checked to c3d and runs a pipeline in Visual3D.

Markerless session:

For markerless session, processing is done in two consecutive steps. In the first step, Theia3D will process all videos and create primary c3d files. In the second step, Visual3D will merge c3d files created by Theia3D with c3d files containing analogue data created by QTM. After merging c3d files processing continues with the same script as for marker-based session.



The processing should be completed with no errors or warnings. If these occur, it usually indicates that the automatic event detection was not successful or that there are some files with missing data. You can use the output in the Pipeline Processing Results to determine which file caused the error/warning. If the problem persists, copy the text from the Pipeline Processing Results into a text file and send them to Qualisys.

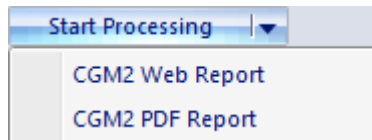


Information about additional report options can be found under “Reports” on page 27.

Note: Session means are calculated as means of within-trial means. In other words, mean of each trial is calculated and those trial means are used to calculate a session mean. Exception are some temporal spacial parameters (stride length, stride width, step length, step time, initial double limb support) where overall session mean is calculated from all steps.

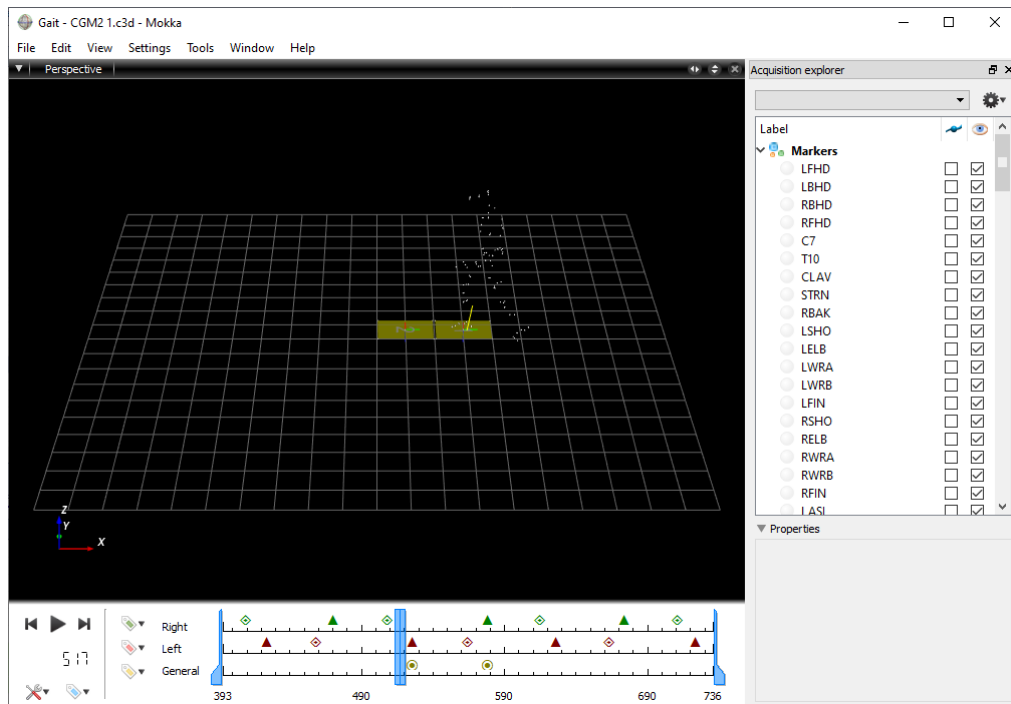
QCGM2

1. QCGM2 processing engine is available only for CGM2 session. When the minimum required number of files has been captured, the Start processing button is available.
2. Click small triangle next to Start Processing button. Two report types are available, online web report and PDF report.



Note: QCGM2 stores .c3d files in 'processed' folder inside working session. Content of this folder is erased each time new processing is started. In order to keep previously edited files from this folder, please store them separately.

3. Select which report you want to be generated. After a while, Mokka software will open, showing first dynamic trial. This step is used to verify foot strike and toe off events. Check that no event is missing and/or no extra events exist. After events are verified, save the changes and close Mokka. Mokka automatically opens for each dynamic trial.



Note that each time QCGM2 processing starts, all previously manually edited event must be set again.

4. After events are verified in all trials, desired report is generated. PDF report is saved as a single pdf file in session directory. If web report is selected, data are uploaded to Report Center and web report is generated in default browser.

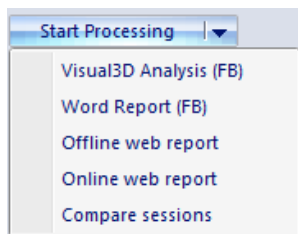
Reports

Overview of report options

When the minimum required number of files has been captured, the Start Processing button is available. Click the small triangle to see the different analysis options:

Note: For CGM2 reporting option see previous chapter.

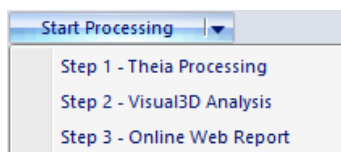
Marker-based session



- a. Visual3D Analysis (default): exports checked files to c3d, runs script in Visual3D, creates Visual3D report but stops analysis before exporting results from Visual3D (Word report is not generated). Make sure you save any changes you make to the Visual3D workspace and close Visual3D before proceeding.
- b. Word Report: opens Visual3D file created in step a., exports results and generates Word report (see “Word report” on page 34).
- c. Offline web report: opens Visual3D file created in step a., exports results and generates offline HTML report (see “Web report” on page 35).
- d. Online web report: opens Visual3D file created in step a., exports results and generates online HTML report (see “Web report” on page 35). Use your qualisys.com credentials to log in to the Report Center to retrieve reports (<https://report.qualisys.com>).
- e. Compare sessions: creates two session comparison report (both in Visual3D and Word), for example for pre-/post-comparisons (see “Comparison report” on page 47). Note that each session must be processed on its own before the data can be compared.

The purpose of a. is that you have the opportunity to check (and modify) the Visual3D analysis before the Word report is generated.

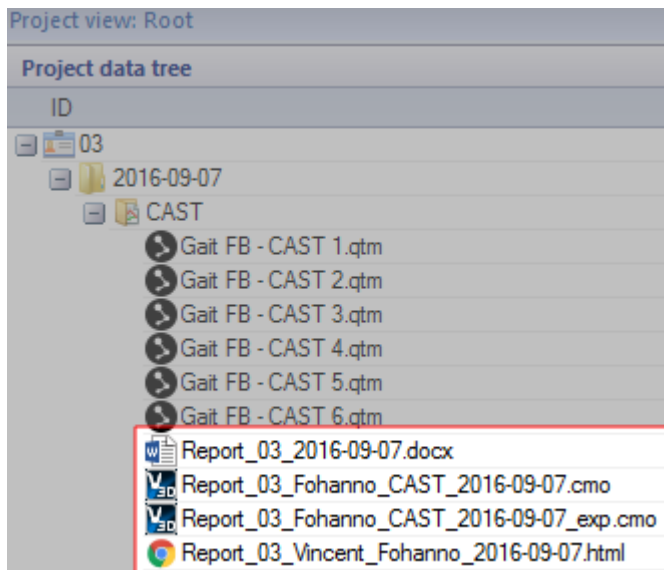
Markerless session



- a. Step 1 - Theia Processing: Starts Theia3D, reads .avi files created by QTM, reorganize them, process all videos and creates c3d file for each trial containing Theia model.
- b. Step 2 - Visual3D Analysis: exports checked .qtm files to c3d, merge them with Theia generated c3d, runs Visual3D scripts and creates Visual3D report.
- c. Step 3 - Online Web Report: opens Visual3D file created in step b., exports results and generates online HTML report (see “Web report” on page 35). Use your qualisys.com credentials to log in to the Report Center to retrieve reports (<https://report.qualisys.com>).

Note: These steps must be run in consecutive order.

To re-open a report at a later stage, double-click its name in the Project data tree of the Project automation pane in QTM:

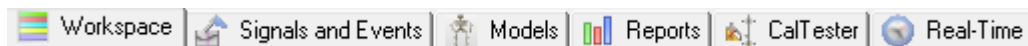


Visual3D Report

The purpose of the Visual3D report is to verify that all data has been processed correctly.

Reviewing data in Visual3D for quality assurance

The Visual3D user interface consists of several tabs:

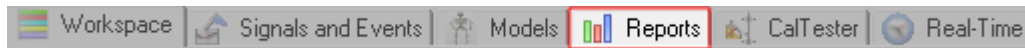


It is best practice to use the following tabs when reviewing for data quality:

- Workspace: overview of all files that are included in the analysis.
- Signals and Events: confirm and edit gait events.
- Models: review the static file and model definition.
- Reports: review processing results.

Report graphs

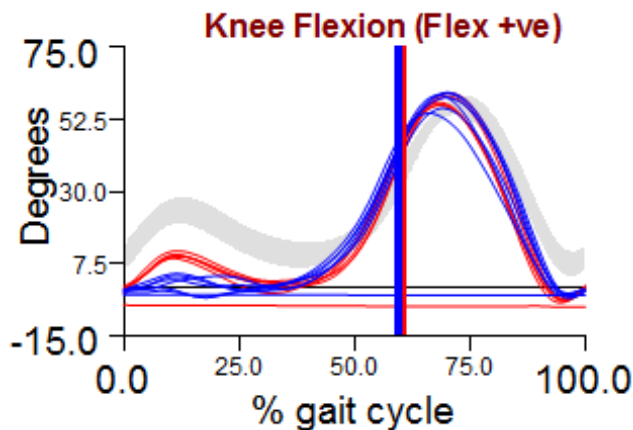
To review the Visual3D report, select the Reports tab:



Each line represents one gait cycle, time-normalised to 0 to 100%. For parameters that are related to the left body side, the left gait cycle is used. Accordingly, the right gait cycle is used for parameters related to the right body side.

The band in the background represents mean and ± 1 standard deviation from the mean of normative data stored at Root\Template\.

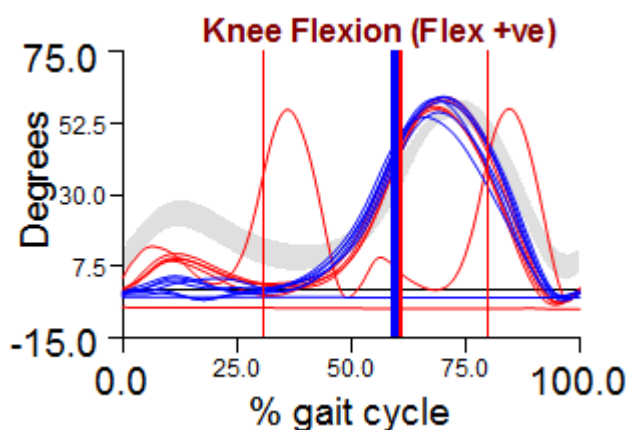
Consistency plots show curves for all trials, so they are useful when verifying data quality:



Notes about the presentation of the consistency plots in Visual3D:

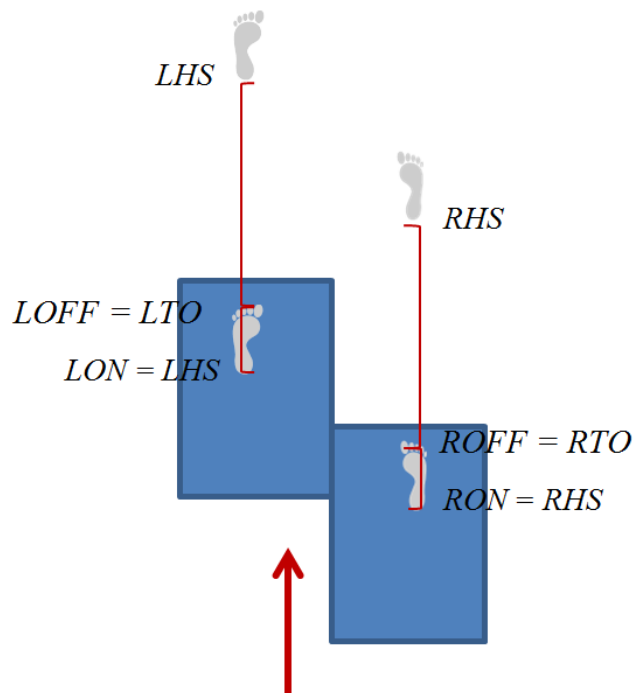
- 0 and 100 % represent heel strike events.
- The vertical line represents the toe off event.
- The horizontal line represents the joint angle in the static trial.

In the following example, an event has been miss-identified. This must be rectified before finalising the report (see “Modifying events in Visual3D” on page 31).



Gait events

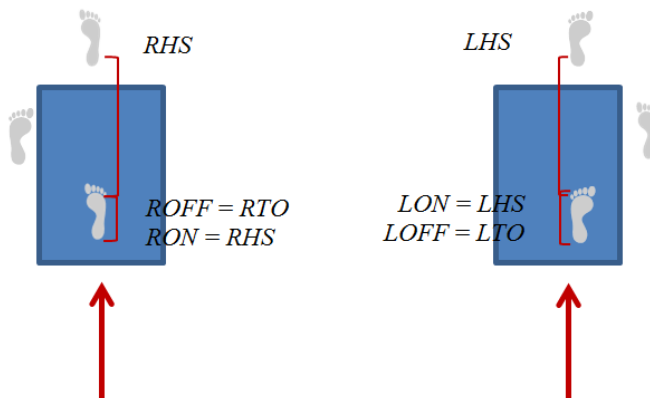
The typical sequence of gait events looks like this for labs with two or more force plates (note: the order of left and right may change depending on the placement of the force plates):



Gait direction

(= positive or negative global/lab X-axis or Y-axis)

For labs with a single force plate, only the left or right foot should make contact with the force plate in each trial:



Gait direction

(= positive or negative global/lab X-axis or Y-axis)

The following tables presents the sequence of events for a left and right gait cycle:

#	Label	Description	Left Stride	Right Stride
	RTO	[removed in event mode "multiple force plates"]		
1	RHS / RON	Right Heel Strike* / Right foot ON FP		Start
2	LTO	Left Toe Off		
3	LHS / LON	Left Heel Strike* / Left foot ON FP	Start	
4	RTO / ROFF	Right Toe Off / Right foot OFF FP		
5	RHS	Right Heel Strike*		End
6	LTO / LOFF	Left Toe Off / Left foot OFF FP		
7	LHS	Left Heel Strike*	End	
8	RTO	Right Toe Off		
	RHS	[removed in event mode "multiple force plates"]		

*Note that LHS/RHS stands for "heel strike" for historical reasons. It is more appropriate to use the term "foot strike" or "initial contact", and the event should be placed at the time of the first contact between foot and ground regardless of which part of the foot makes contact with the ground.

Modifying events in Visual3D

To modify events in Visual3D, go to the Signals and Events tab:



Use the file selector to choose the file you want to review:



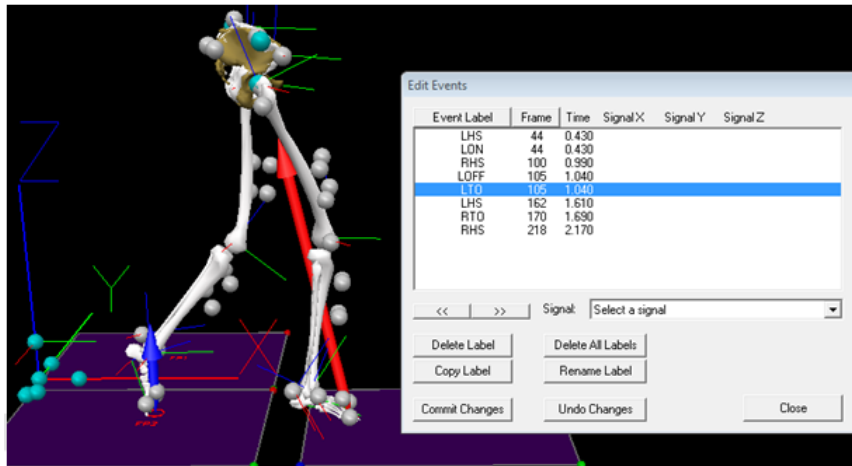
Click the Add events icon to add missing events:



Click the Edit events icon to modify existing events:



In the Edit Events dialog, use the << and >> buttons to move existing events, Delete label to delete an event and Commit Changes when done.



When you are finished, click the Recalc icon to update the report:



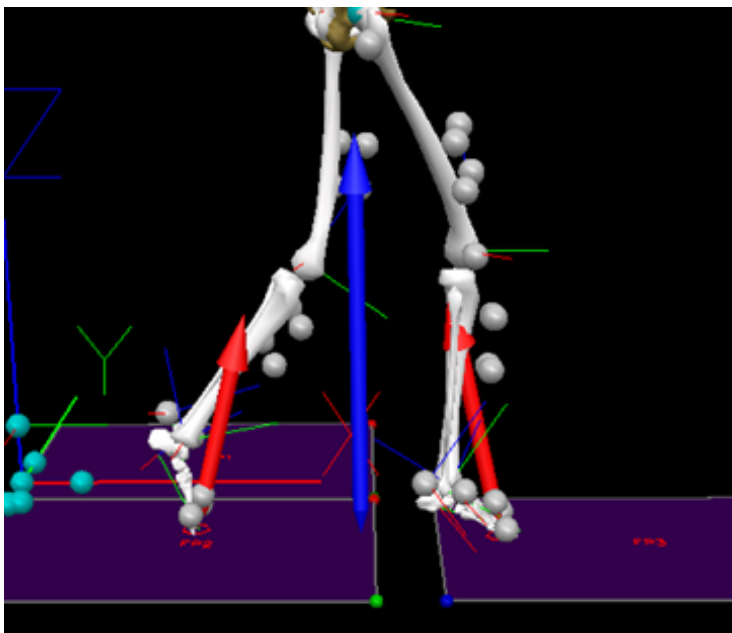
Save the workspace in Visual3D to make sure the changes will also appear in the Word report.

Note: GRF graphs seen in report are not part of Recalc pipeline therefore changes to force assignment will not take an effect on the graphs .

Modifying force assignments in Visual3D

Whenever Visual3D detects that a body segment is making contact with a force plate, the force vector is shown in blue instead of red. It is important to make sure that the assignments are correct, because wrong force assignments result in invalid kinetic results.

The following screenshot shows an example for an incorrect force assignment where Visual3D miss-identified which foot made contact with the second force plate:



Recalc Pipeline EVENT EVENT RESET F F F F Gait FB - CAST 2.c3d

View / Modify Force Platform Segment Assignments

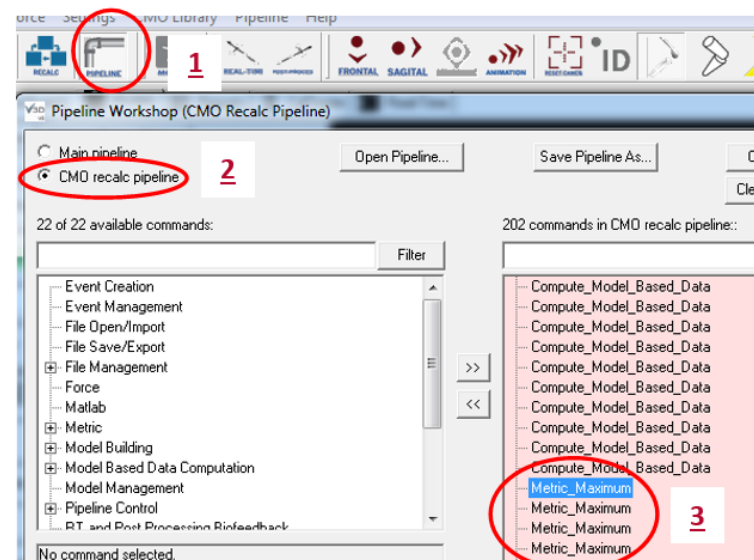
Force Item	Segment	Start Frame	End Frame	Modified fr...	Original Se...	Original
FP2	Left Foot	41	112	No		
FP3	Left Foot	99	223	No		

111

Add Segment Assignment Modify Segment Assignment Delete Segment Assignment

OK

To avoid pipeline processing warnings some commands need to be removed from the Recalc pipeline before saving the custom report template. To do so, open custom template in Visual3D, (1) open Pipeline Workshop, (2) click CMO recalc pipeline and (3) remove all commands below last `Compute_Model_Based_Data` command. Then save the report template.

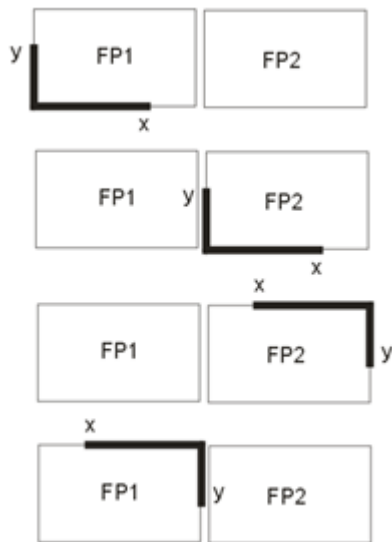


Centre of Pressure plots (COP)

COP plots require either one force plate, or two force plates. More than two force plates are not supported. If a laboratory contains more than two force plates those included in analysis must be FP1 and FP2 for two FP setup or FP1 for single FP setup. Following picture shows supported L-frame position in respect to FPs. Please contact Qualisys support if walking in Y direction is required.

COP graphs are included in Visual3D report only.

To switch off COP pages from Visual3D report open Template/Settings.php, scroll to `$show_COP = true`; and change it to `$show_COP = false`.



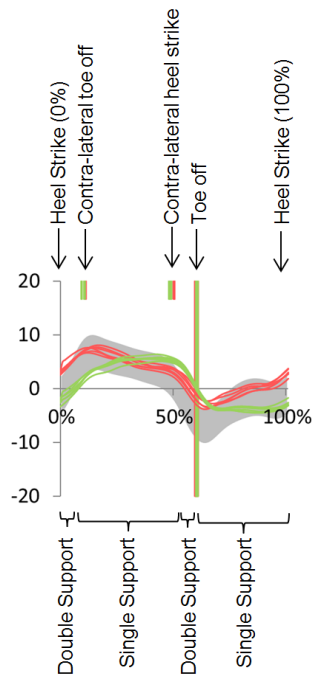
Word report

Graphs

Lines in consistency plots represent within-trial mean values, i.e. there is one line for each trial (default colours: blue for right side and red for left body side. To modify colors of line graphs, please open Template/settings.php and edit values within the “//Word report graphs settings” section.). The grey band represents ± 1 standard deviation from the mean normative value.

Graph annotations

The following figure shows the meaning of the graph annotations in the Word report.



Word report customisations

The appearance of the curves in the graphs can be changed by modifying the file `Templates/settings.php` with a text editor. The file includes a section with "Word report settings". Each line includes a description of the setting and the available option, such as line colors and whether all trials are plotted or individual means.

The general layout (logos etc.) can be edited by opening `Templates\Assets\Word Templates\Report_template.docx`. Overwrite the existing file after making the changes.

Note: While saving custom Word template it is very important to save it in compatibility mode. To do so make sure that in "Save As" dialog checker "Maintain compatibility with previous versions of Word" is checked before saving edited document. Failing to do so will significantly slow editing and scrolling the Word document.

Word graph layout can be changed at `Templates\Assets\Word Templates\graph_templates.xlsx`. Save modified template (.crtx) under same name at Template folder. It is advised to create a backup of existing crtx files before overwriting them.

Web report

Introduction

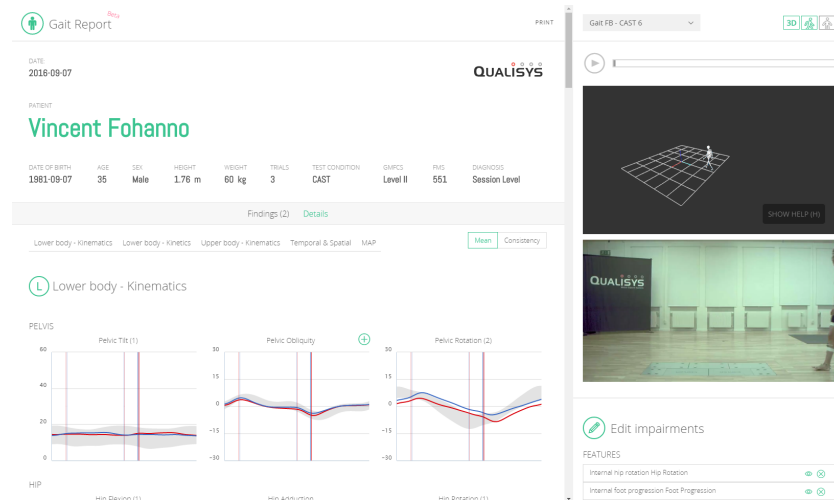
The web report is an interactive report displaying analysis results, a 3D viewer with "skeleton" and video players.

Before generating a web report, the data must be processed with Visual3D, using the Visual3D Analysis option in the *Start processing* dropdown menu (QTM > Project view).

If you select Offline web report, the report will be stored locally on your computer and cannot be accessed by any external person, unless you share it by passing the .html file on to someone. If you select Online web report,

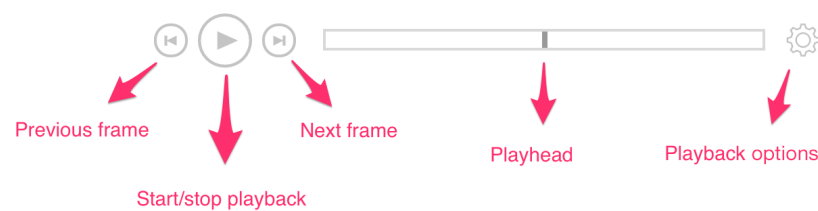
the data is uploaded to a secure cloud storage location. Use your qualisys.com credentials to log in to the Report Center to retrieve, edit or share reports (<https://report.qualisys.com>).

The report is separated into two panes: the main pane (left) displays Details and Findings. The interactive pane (right) contains 3D viewer, video players and the Edit impairments section.



Playback

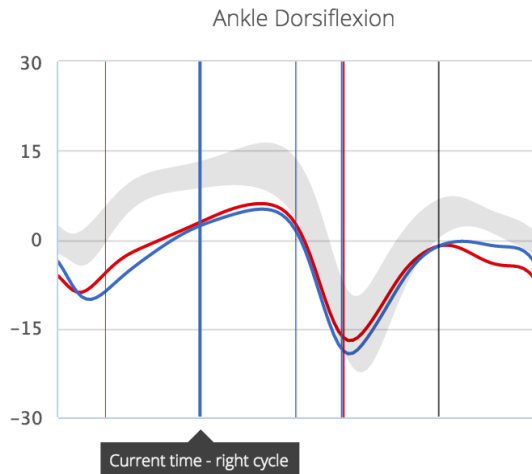
Video and 3D viewer playback is controlled by the playback controls:



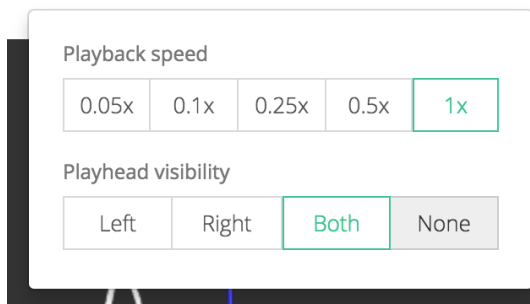
To move forward or backward within the file, click and move the playhead while holding the mouse button. You can also step forward or backward one frame at a time with the previous/next frame buttons or by using the left (←) and right (→) shortcut keys.

The current time is indicated by the timeline playhead and by thin grey vertical lines in timeseries charts. These lines, or playheads, represent the current time relative to individual gait cycles. There is one playhead for the left cycles and one for the right. Note that two playheads are needed since the left and right gait cycles have a temporal offset.

In consistency mode, the current cycle gets highlighted as the video and 3D viewer playback updates.



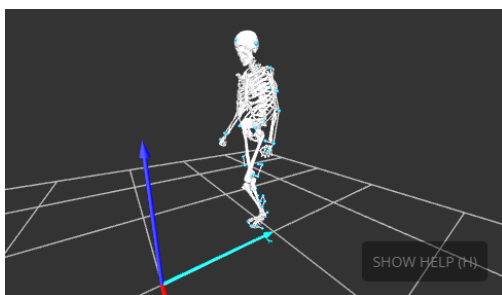
You can set the visibility of the chart playheads individually via the playback options (shown below). You can also change the playback speed.



The 3D viewer and video players can be hidden by toggling the corresponding buttons:



3D viewer

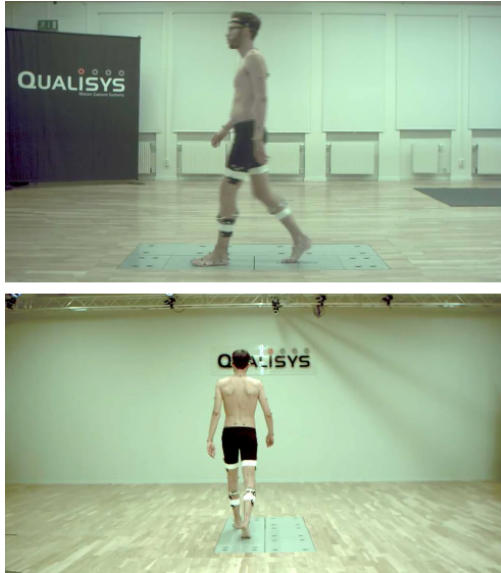


The 3D viewer shows the markers belonging to the model and the body segments (skeleton).

- Rotate by holding the left mouse button.
- Pan is disabled as the camera automatically follows the body model.
- Zoom by using the scroll wheel or middle mouse button.

Video

The report can show up to two videos per measurement. Typically you want one sagittal and one frontal camera. Note that the correct suffix for the recorded video files must be specified prior to report generation, see “Video setup for web report” on page 10. The video included in the report are compressed versions of the original captures. The compression takes place when generating the report.



Edit impairments

In the Edit impairments section of the interactive pane, you can add and edit impairments. See “Workflow for adding features and impairments” on page 46 for more details.



Edit impairments

FEATURES

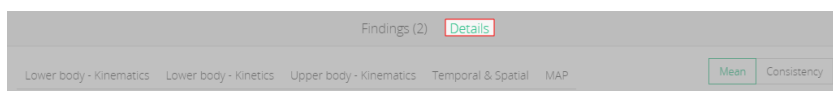
Internal hip rotation Hip Rotation	 
Internal foot progression Foot Progression	 
Decreased anterior tilt throughout Pelvic Tilt	 
Restricted hip flexion in early stance Hip Flexion	 
Knee extension within normal limits in late stance Knee Flexion	 

Enter impairment

 Add impairment

Details

The details pane shows charts for the kinematic and kinetic parameters, tempo-spatial parameters, the Movement Analysis Profile (MAP) and EMG.

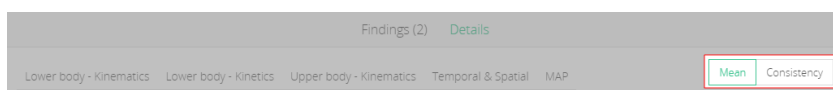


Then navigation bar can be used to quickly move from one group of parameters to another:



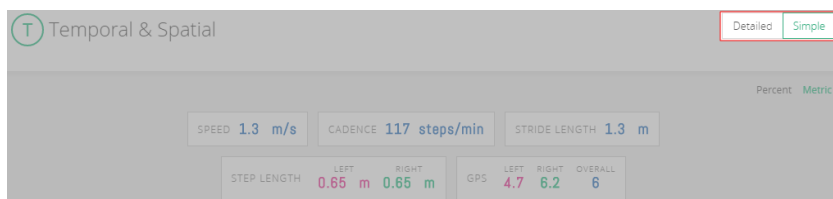
Kinematic and kinetic timeseries

Use the mean/consistency buttons to set whether the charts show the mean of all trials or all individual curves (Consistency):



Temporal & spatial parameters

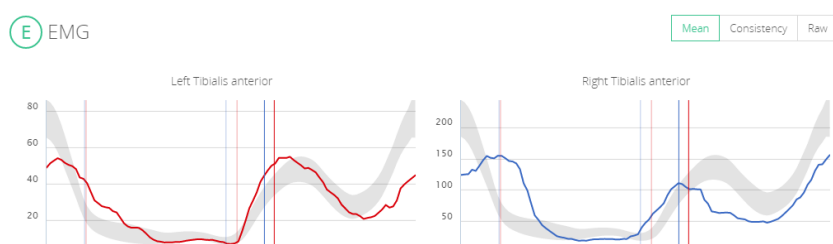
The tempo-spatial parameters can be displayed in simple view (numeric values only) or detailed (bar charts):



You can further display the simple view as metric values or percentages. Percentages are normalised to leg length and expressed as % of the normative values (see “Appendix B - Normative data” on page 54 for how to set your own normatives).

EMG data

The web report can include EMG data that can be displayed as processed EMG (mean, consistency) or raw EMG.



First, make sure that your EMG channels adheres to the [EMG naming conventions](#).

Then, all EMG channels defined in QTM will be plotted in the EMG section of the report (Details tab).

Physical examination

You can insert physical examination data in the *Physical Examination* tab. Once a physical examination has been entered, you can reference individual examination fields when you define impairments. For more information on how to associate physical examination fields with impairments, see “Impairments” on page 44.

Inserting physical examination data

To add physical examination data simply type in the information using the input fields. Each row corresponds to a parameter with one or two values. As you type, new rows are automatically added.

Note that instead of typing manually, you can also copy and paste a table directly from Microsoft Excel. Please note that when pasting from Excel, only the first three columns will be inserted.

Findings (1) Details **Physical Examination**

 Edit physical examination

Hip flexor strength	4+(2)	4+(2)
Hip extensor strength (knee 0°)	4+(2)	4(2)
Enter supplementary data	Left	Right

🔗 You can copy cells and rows from an Excel spreadsheet and paste it into the clinical results.

Attachments

It can sometimes be useful to include files generated by other software in a report. To do this, you can use attachments.

Findings (1) Details Physical Examination **Attachments**

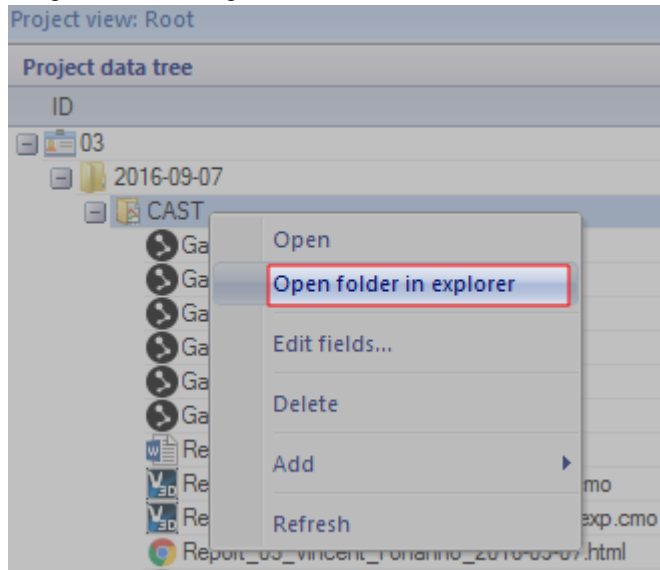
 Attached files


Physical examination.xlsx


Journal.pdf

Attaching files to a report

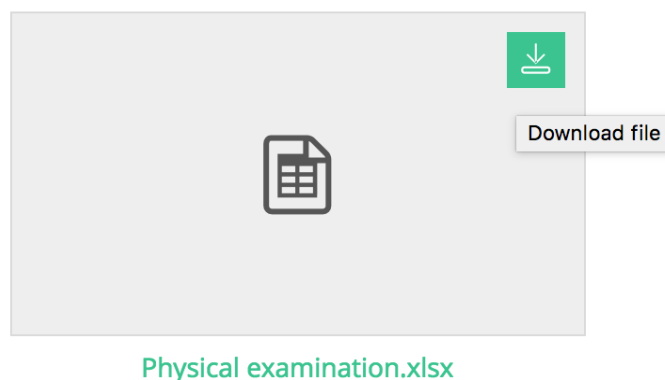
1. Open the data folder of your current session. In QTM, go to Project view > Right click measurement > Open folder in explorer.



2. In Explorer, create an empty folder and name it 'attachments' (it should reside next to the .qtm files of your session folder).
3. Place one or more files in the new folder. These files will now be embedded in the next report you generate.

Viewing attachments in a report

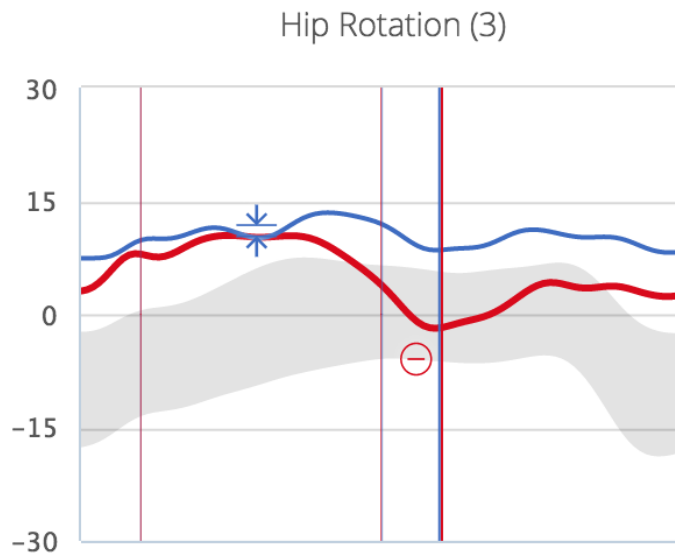
1. Go to the Attachments tab.
2. Your attachments can now be viewed or downloaded.
 1. To view an attachment, simply click on it.
 2. To download an attachment, click on the download icon in the top right corner of the attachment you want to download.



Gait features

Gait features are used to annotate charts with relevant information. You can add features with or without symbols. Symbols should be used when you want to annotate a specific part of a chart while a chart-wide feature is used to make a note associated with the entire chart.

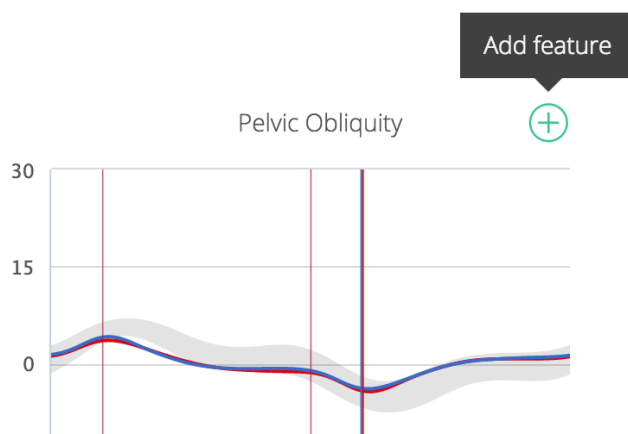
Note that you cannot create impairments without first creating at least one feature.



The number in brackets in the chart title indicates the number of features that have already been added to a chart. See “Workflow for adding features and impairments” on page 46 for details.

Adding a feature without symbol

1. Double-click anywhere on the chart or click the (+) button to the left of the chart title.



2. Make sure that 'No icon' is selected.

3. Edit the description of the feature as needed. You can also write a comment.
4. Save the feature by clicking Save or use the Shift+Enter keyboard shortcut.

Adding a feature with symbol

1. Double-click on the chart where you want to add a feature symbol.
2. Select the symbol you want to use. A description of the feature is automatically generated.
3. Edit the description of the feature as needed. You can also write a comment.
4. Select which curve(s) the feature relates to (left, right, bilateral).
5. Save the feature by clicking Save or use the Shift+Enter keyboard shortcut.

Editing a feature

To edit a feature either double-click a symbol in a chart or click on the edit icon  in the features list.

Tip: to scroll to the chart that contains a feature, click on the eye icon .

Reordering features

You can reorder features in the feature list of the *Edit impairments* section.

1. Drag the feature you want to reorder.
2. Drop the feature in the wanted position.

FEATURES

	Abnormal slope of left pelvic rotation	Pelvic Rotation	
	Not enough left hip rotation (1)	Hip Rotation	
	Decreased range of right hip rotation (1)	Hip Rotation	  
	Too much bilateral hip adduction	Hip Adduction	
	Bilateral hip flexion within normal limits	Hip Flexion	

Impairments

Gait features can be associated with one or more *impairments*. In addition to features, an impairment can also be associated with supplementary data.

Creating impairments

1. Click on one or more features to select them in the feature list of the *Edit impairments* section.
2. Give the impairment a descriptive name, and click the *Add impairment* button. The newly created impairment will show up at the bottom of the impairment list of the *Edit impairments* section

To add supplementary data or set the Evidence and Effect on walking levels, see “Editing impairments” on the next page.

FEATURES

	Abnormal slope of left pelvic rotation	Pelvic Rotation	
	Not enough left hip rotation	Hip Rotation	  
	Decreased range of right hip rotation	Hip Rotation	
	Bilateral hip flexion within normal limits	Hip Flexion	
	Too much bilateral hip adduction	Hip Adduction	

My impairment

 Add impairment

Editing impairments

In the impairments list of the *Edit impairments* section:

1. Click to expand the impairment you want to edit.
2. Now you can:
 - i. Use the dropdown menus to set the Evidence and Effect on walking levels.
 - ii. Remove a feature from the impairment by hovering the feature and clicking the trash can icon next to it.
 - iii. Add a feature to the impairment by dragging and dropping features from the feature list onto the impairment.
 - iv. Add supplementary data. Either by typing them in or copy/pasting from Microsoft Excel. When typing, you will get auto-complete suggestions from the physical examination if one exists. See *Inserting a physical examination* for details.

IMPAIRMENTS

My impairment (2) 

EVIDENCE:

EFFECT ON WALKING:

Features

⊖

 Not enough left hip rotation (Hip Rotation)

✱

 Decreased range of right hip rotation (Hip Rotation)

Supplementary data

Enter supplement

L

R

Comment

Removing impairments

In the impairments list of the *Edit impairments* section:

1. Click to trash can icon to the right of the impairment name.

Findings

The findings tab is intended for users of the report that require an overview of the findings after the analysis has taken place and been processed. The structure is based on a template prepared by Baker, McGinley and Thomason for the workshop 'Impairment focused interpretation' at the JEGM meeting, Miami, 2010. Some adaptations have been made based on other example reports (Baker, 2013). The following sections are included:

- Summary
- Orientation
- Data quality
- Evidence and Interpretation, see “Workflow for adding features and impairments” on the next page.

Workflow for adding features and impairments

1. Quality assurance: this must take place before any interpretations is done. For example, pathological gait patterns often result in erroneous events that must be corrected before the web report is generated, see “Reviewing data in Visual3D for quality assurance” on page 28. Consistency plots must be reviewed at this stage. If there are outliers, it is recommended to review the original files to ensure that all markers are labelled correctly
2. Mark-up: this is the process of identifying features in the gait charts.

A form with two input fields: "Enter feature" and "Enter comment". Below the "Enter feature" field is a label "Save with Shift+Enter". At the bottom right are two buttons: "Cancel" and "Save".

3. Grouping: this is the process of grouping features and supplementary data and associating them with specific impairments. This step is done in the Edit impairments section in the interactive pane (left), below the 3D/video viewers. For details, see the “Impairments” on page 44 section.
4. Reporting: documentation of the interpretation process. The Findings pane includes text boxes to add additional information for the referrer.

Entries to these text boxes are automatically saved locally inside the web browser. This means that they will be preserved on the local computer (e.g. when re-generating the report), but to share the comments with other users, the report much be printed using the Print feature.

A form with two main sections: "HISTORY" and "COMMENTS ON VIDEO". The "HISTORY" section has a label "Notes on relevant events in patient's history" and a text area containing a detailed report. The "COMMENTS ON VIDEO" section has a label "Comment" and a text area containing a detailed comment. Below these sections is a "Data quality" section with a green checkmark icon and a label "Data quality". It contains four sub-sections: "REPRESENTATIVITY", "CONSISTENCY", "DATA ARTEFACTS", and "Is the data likely to be representative of the subject's usual walking pattern?". Each sub-section has a text area for input.

More detailed information about impairment focused interpretation can be found in Baker (2013), chapter 12.

Customising colours

The colour of the curves in the charts can be changed by modifying the file Templates/settings.php with a text editor. The file includes a section with "General" report settings, including \$line_color_left and \$line_color_right. By default, red is used for the left body side and blue for the right body side.

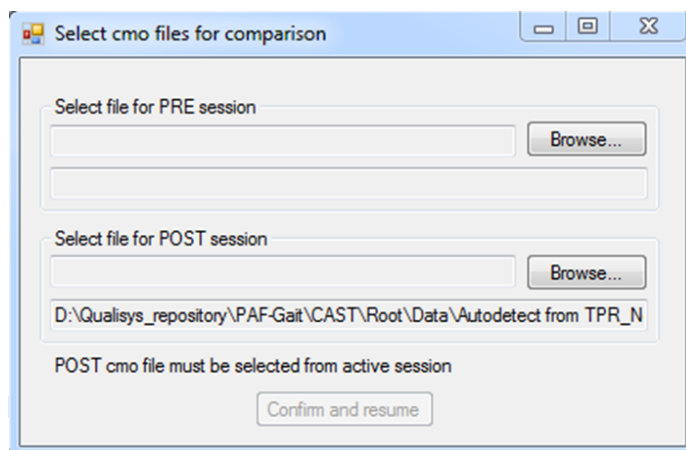
Comparison report

The comparison report allows to compare two different sessions. It is assumed that sessions being compared have same attributes, e.g. full body/lower body, same EMG signals, force plate, etc. Both Visual3D and Word report show comparison data.

Note: more comprehensive comparisons of up to five sessions and PDF export are available within online web reports.

Follow these steps to generate an offline comparison report:

1. Process the first ("pre-") session with "Analysis only", check data, then use "Report only" to create a Word report.
2. Process the second ("post-") session with "Analysis only", check data, then use "Report only" to create a Word report.
3. Now click "Compare sessions" in the "post" session. Make sure you select the first session as "pre-", and the session where you clicked "Analysis" as "post-session":



By clicking on Browse... select *.exp.r3w files for PRE and POST session and then click Confirm and resume.

Additional notes:

- The file *.exp.r3w must exist for both sessions before generating comparison report.
- Visual3D comparison report includes all pages, even empty ones. Word comparison report removes pages that are not relevant.
- To modify colours of line graphs, please open settings.php and edit values at “//Word report graphs settings” section.

Special analysis conditions

Inclined treadmill

It is possible to use the module with inclining treadmills. The following points must be considered:

- Marker placement: four markers must be placed symmetrically to the left and right of the belt as shown (for AMTI instrumented treadmills, use the existing threads):



- The markers must be labelled as shown in the picture in all trials (including static trial). To do so, it is recommended to create a separate AIM model containing only the treadmill markers (for example treadmill.qam).

By default, only the standard running AIM model will be applied to the trials at the time of capture. To label the treadmill markers, wait until you have finished the data collection session and use Batch processing to apply two AIM models to all trials (Running.qam and Treadmill.qam). See the QTM manual for details on how to create and apply AIM models.

If you experience difficulties with the AIM model due to symmetry (i.e. left/right or front/back swaps), add a fifth marker on the treadmill to introduce asymmetry. The marker can have any name as it is not used in the analysis but must be included in the AIM model.

- If the markers are present and labelled correctly, the position of the treadmill belt will be automatically adjusted in Visual3D for each trial based on the position of the markers.
- During calibration, no incline should be applied (L-frame must be in horizontal plane).
- If the treadmill is instrumented with force plates, see “Instrumented treadmill” below. If the treadmill is not instrumented, the event mode "No force plates (automatic)" is usually the best choice for the gait event detection.
- Make sure you enter the correct treadmill speed in the trial details in QTM for each recorded trial.

Instrumented treadmill

It is possible to use the module with instrumented treadmills. The following points must be considered:

- Set the event mode in the session details to "Instrumented treadmill (split belt)" or "Instrumented treadmill (single belt)", depending on the type of treadmill you are using.
- Enter the correct treadmill speed in the trial details in QTM.
- Visual3D automatically assigns the force to the left/right foot depending on which foot is in contact with the treadmill. The force vector will appear blue in Visual3D to indicate that it has been assigned to a body segment.
- Both feet must be on separate belts of the force plates at all times. If both feet make contact with the same belt at some point, the file needs to be cropped to exclude these sections.
- For single belt treadmill double support phase is excluded from the analysis.

Appendix A - EMG

Introduction

Gait Module automatically detects EMG signals collected by Noraxon, Delsys Trigno and Cometa/Myon devices if the data is collected in QTM via a digital integration. In case Board name is set to Noraxon or Delsys Trigno, EMG signals are processed and added to both Visual3D and Word report. EMG data that is collected via a Qualisys analog board is also supported (for details see naming conventions below).

If you are using one of the digital integrations, please refer to the QTM manual (Appendix F) for details on how to install the EMG device and to set up the connection.

Please follow the manufacturers' instructions when attaching the sensors and check the signal quality prior to data collection.

Naming convention for EMG channels

The EMG data will only appear in the report when the EMG names comply with the following rules:

1. Each name starts with body side identification. Only “Right“ or Left” or “R” or “L” followed by “single space” or “underscore” followed by signal name is recognized by the module. For example “Right Biceps Femoris” or “L_Gluteus Maximus”.

2. The signal names must match exactly one of the following labels:

Biceps Femoris

Erector Spinae lumbar

Erector Spinae thoracic

External Oblique

Gastrocnemius lateral

Gastrocnemius medial

Gluteus Maximus

Gluteus Medius

Hip Adductors

Internal Oblique

Multifidus

Peroneus

Rectus Abdominis
Rectus Femoris
Semitendinosus
Soleus
Tensor Fasciae Latae
Tibialis Anterior
Trapezius upper
Vastus Lateralis
Vastus Medialis

3. Signals will be plotted in the same order as entered to device setup.
4. For EMG data that is collected via a Qualisys AD board (USB-2533) the prefix “EMG_” must be added to all EMG signals.

There are several options to change the way EMG signals are presented in Word report. To do so open settings.php from the Templates folder, scroll down to //Word report graphs settings and follow the instructions in the file:

1. The colour of the EMG signals in the Word report line graphs can be changed if required (only red, green, blue, purple, orange and black).
2. It is possible to plot each trial or mean of trials.
3. By default processed and raw EMG signals are plotted on separate page. It is possible to combine them into one graph

Note: Noraxon normative data are stored as an encrypted file in [Project folder]\Templates\EMG_normatives_Noraxon.vnd.encr. The file is only used if a Noraxon device has been used to collect the data directly in QTM via the digital integration. Non-Noraxon users can store their own normative data in Root\Template\EMG_normatives.vnd. In absence of EMG_normatives.vnd there will be a number of warnings and errors that can be switch-off by editing Settings.php at Template folder. Find \$hide_messages = false; and change it to \$hide_messages = true;

Note: Incorrect signal names will result in error messages in Pipeline Processing Results and normative data for EMG signals cannot be assigned to graphs.

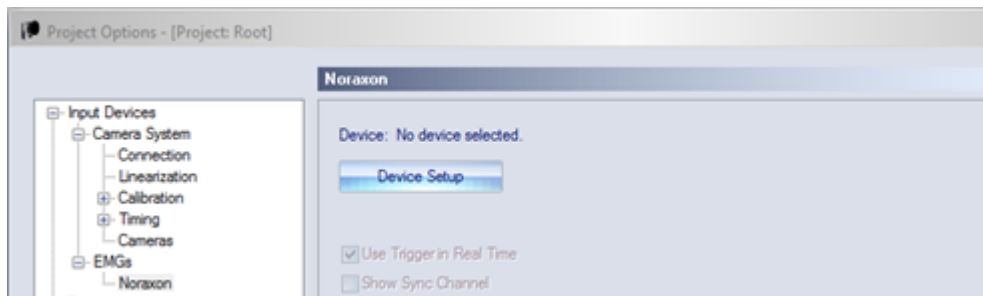
Note: EMG signals are added to the reports dynamically based on the EMG channels found in the QTM files; therefore they are not included in the report template and cannot be modified by changing the report template in Visual3D.

How to set up channel names

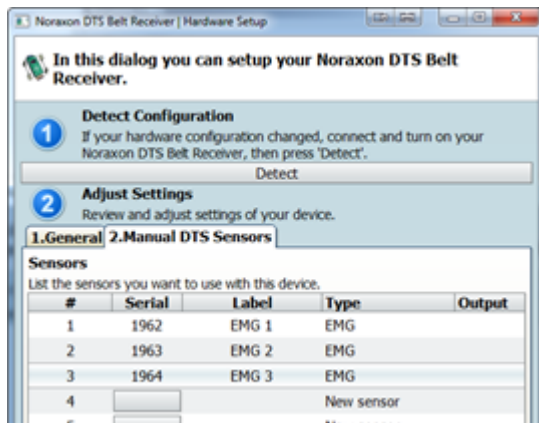
The following sections describe how to adapt the channel names to the naming convention for the EMG systems that are supported by the Gait module report.

Noraxon

In Project options click on Noraxon EMG, then click on Device setup and open Noraxon Hardware Setup dialog.



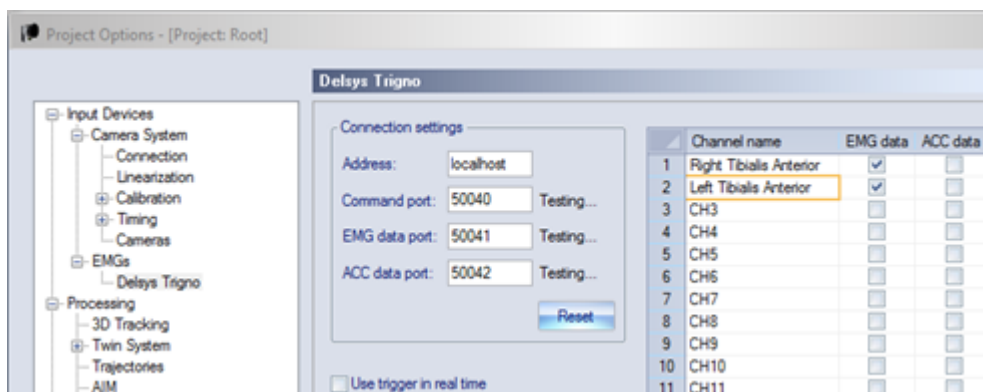
In Noraxon Hardware Setup assign appropriate EMG signal names (Label) to each serial number (Serial). See above for naming convention.



It is possible to use hardware profiles to store the names for different types of data collection (e.g. uni-lateral, bi-lateral).

Delsys Trigno/Avanti

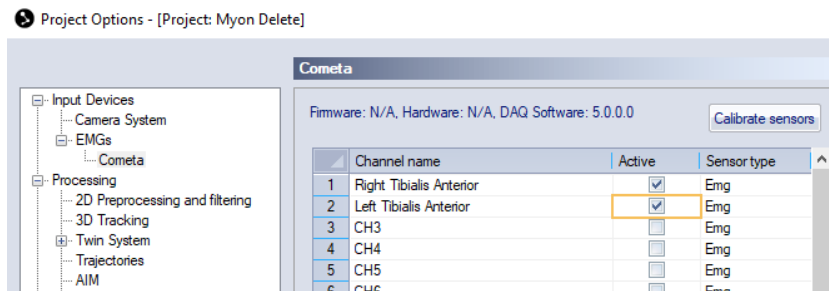
Enter the muscle names under Channel name. See above for the naming convention.



Note: Acceleration signals are ignored and not displayed in the reports.

Myon / Cometa

Enter the muscle names under Channel name. See above for the naming convention.



Note: Acceleration signals are ignored and not displayed in the reports.

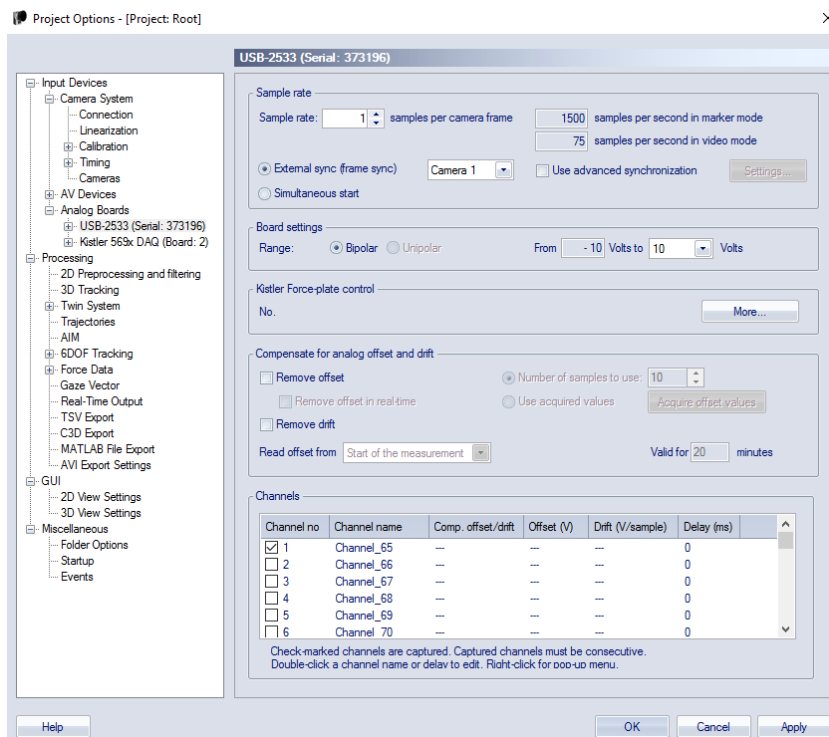
Analog board

Enter the muscle names by changing the analog channel names in Project Options > Input Devices > Analog boards > USB-2533. See [EMG naming conventions](#) for correct signal names.

Important: All EMG channels must be prefixed by "EMG_" so that the channels is recognised as EMG channel during processing.

Only the USB-2533 board is supported.

Important: you must enter the correct scaling factor from the analog data (V) to the original EMG units (microV) in the file [project folder]\Templates\settings.php. Open the file in a text editor and enter the value. Refer to the EMG manufacturer's manual for the correct factor. The default factor is 1000.



Appendix B - Normative data

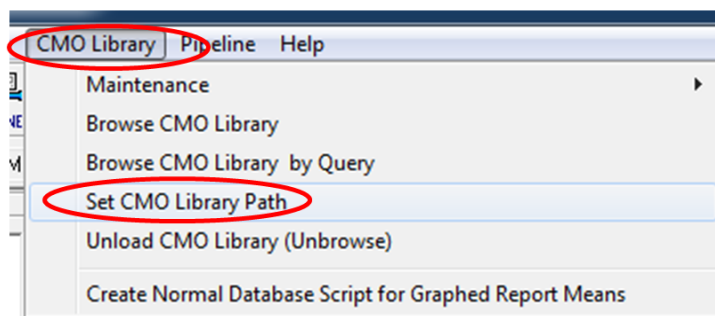
It is good practice to establish a normative data set for your lab. This ensures that it reflects your normal procedures (e.g. placement of anatomical markers) and lab environment (e.g. gait speed depending on lab size). This section describes how to update the normative database.

1. Select .cmo files which you want to create the normative data with.

Copy them into one dedicated folder (e.g. D:\Qualisys_data\Gait_data\Normatives). It is assumed that .cmo files were generated by Gait module and include all needed information (correct Tags, signals, etc.)

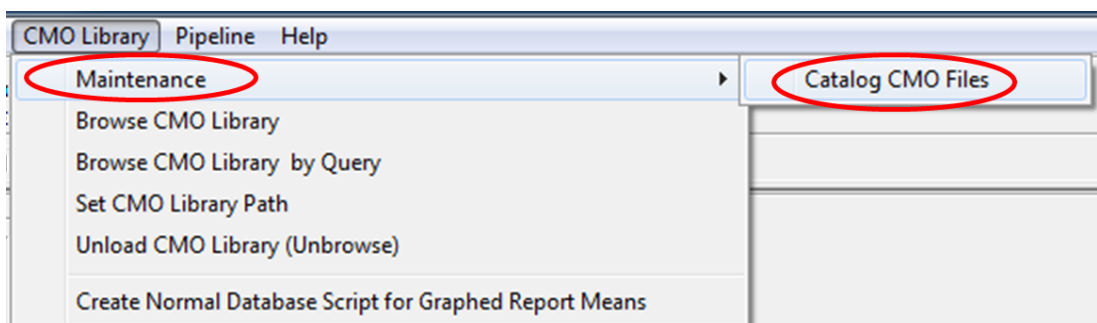
2. Set path to CMO Library.

Open Visual3D, open “CMO Library” > “Set CMO library Path”, then set path to folder that you created for the .cmo files.



3. Catalog files in CMO library.

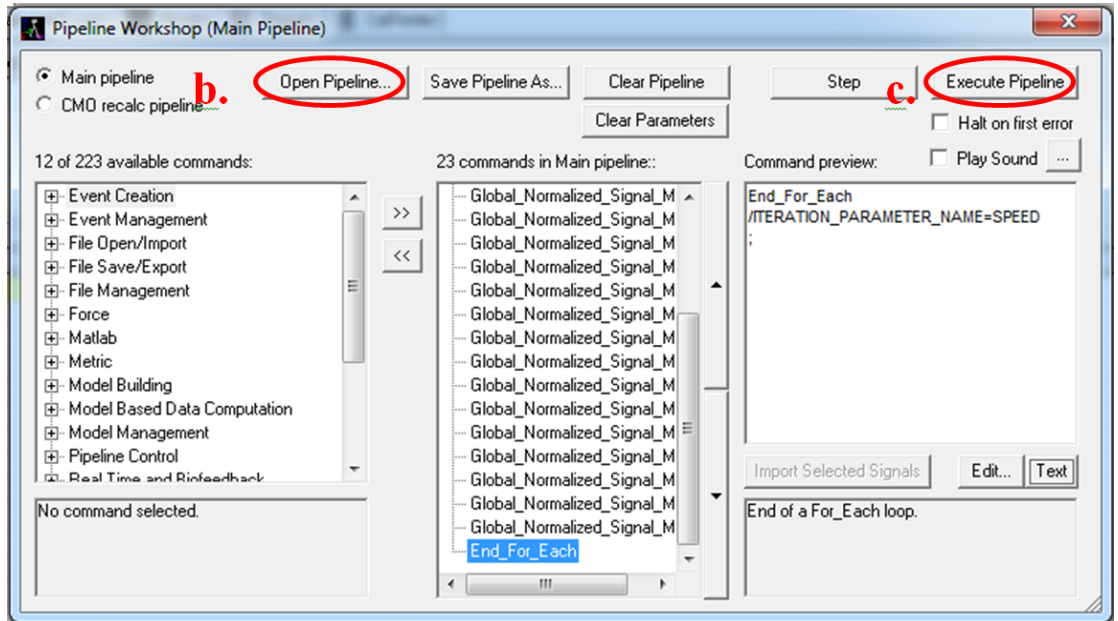
Open “CMO Library” > “Maintenance” > “Catalog CMO Files”.



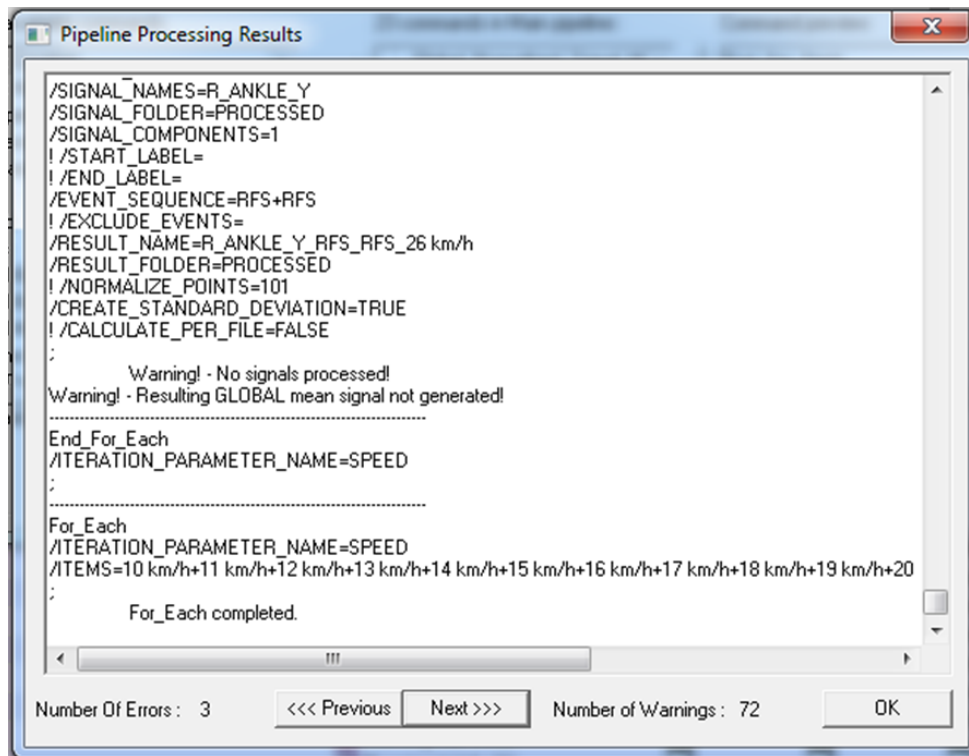
4. Run the script “normatives.v3s”:
 - a. In Visual3D click on Pipeline icon below the menu bar and a “Pipeline workshop” window will open.

- b. Click “Open Pipeline...” and load `normatives.v3s`. It is located in the folder “[Your QTM project]\Templates\Scripts\src\Visual3D”.
- c. Click “Execute Pipeline”.

Visual3D will prompt for path to `.cmo` files, then ask if normative sample also includes upper body signals and Oxford foot model signals.

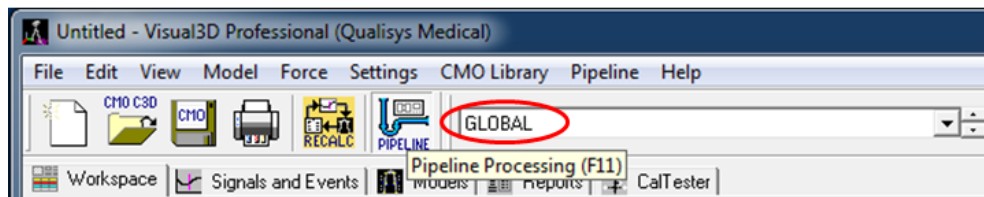


5. A “Pipeline Processing Results” window will pop-up after completion of the script. Check for errors and warnings.

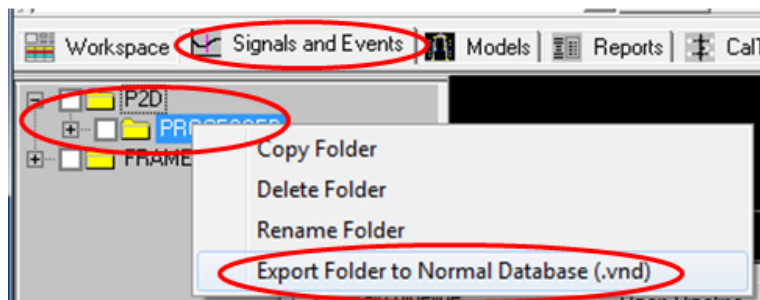


6. Set workspace to Global.

After completion of normatives.v3s script, set the left File Selection Combo Box to "GLOBAL".



7. Export normatives into .vnd files:



- a. Export timeseries normatives:

Right click on P2D > PROCESSED. Name file "normal_database.vnd" and save it to Templates folder of the analysis module project.

It is recommended to back up the previous version of .vnd files.

8. The new normative data will be used the next time a report is generated with Gait module.

Note: please contact Qualisys Support if you would like to update the normative data that is displayed in the online web report.

Appendix C - Joint angles

General rotation sequence is X, Y, Z.

Exception: absolute angles relative to the lab (Pelvis Angle, Thorax-Lab Angle) use Z, Y, X (Baker, 2001, Gait and Posture 13).

Joint	Segment/ Reference	Negate?		Joint	Segment/ Reference	Negate?
Left Shoulder Angles	LAR/RTA	X=FALSE Y=TRUE Z=TRUE		Right Shoulder Angles	RAR/RTA	X=FALSE Y=FALSE Z=FALSE
Left Elbow Angles	LFA/LAR	X=FALSE Y=TRUE Z=TRUE		Right Elbow Angles	RFA/RAR	X=FALSE Y=FALSE Z=FALSE
Left Thorax Angles	RTA/RPV	X=TRUE Y=FALSE Z=TRUE		Right Thorax Angles	RTA/RPV	X=TRUE Y=TRUE Z=FALSE
Left Thorax Lab Angles*	RTA/Virtual Lab	X=TRUE Y=FALSE Z=TRUE		Right Thorax Lab Angles*	RTA/Virtual Lab	X=TRUE Y=TRUE Z=FALSE
Left Pelvic Angles*	RPV/Virtual Lab	X=TRUE Y=FALSE Z=TRUE		Right Pelvic Angles*	RPV/ Virtual Lab	X=TRUE Y=TRUE Z=FALSE
Left Hip Angles	LTH/RPV	X=FALSE Y=TRUE Z=TRUE		Right Hip Angles	RTH/RPV	X=FALSE Y=FALSE Z=FALSE
Left Knee Angles	LSK/LTH	X=TRUE Y=TRUE Z=TRUE		Right Knee Angles	RSK/RTH	X=TRUE Y=FALSE Z=FALSE
Left Ankle Angles	LMF/LSK**	X=FALSE Y=TRUE Z=TRUE		Right Ankle Angles	RMF/RSK**	X=FALSE Y=FALSE Z=FALSE
Left Foot Progression	LMF/ Virtual Lab	X=FALSE Y=FALSE Z=TRUE		Right Foot Pro- gression	RMF/ Virtual Lab	X=FALSE Y=FALSE Z=FALSE

*Uses Z,Y,X rotation sequence.

**If session option "normalise to static trial" is activated, the segment Left Foot Normalized is used instead of LMF.

Joint	Segment	Resolution coordinate system	Negate?	Joint	Segment	Resolution coordinate system	Negate?
Left Hip Moment	LTH	RPV	X=TRUE Y=FALSE Z=FALSE	Right Hip Moment	RTH	RPV	X=TRUE Y=TRUE Z=TRUE
Left Knee Moment	LSK	LTH	X=FALSE Y=FALSE Z=FALSE	Right Knee Moment	RSK	RTH	X=FALSE Y=TRUE Z=
Left Ankle Moment	LFT	LSK	X=TRUE Y=FALSE Z=FALSE	Right Ankle Moment	RFT	RSK	X=TRUE Y=TRUE Z=TRUE
Left Hip Power	LTH	RPV	X=FALSE Y=FALSE Z=FALSE	Right Hip Power	RTH	RPV	X=FALSE Y=FALSE Z=FALSE
Left Knee Power	LSK	LTH	X=FALSE Y=FALSE Z=FALSE	Right Knee Power	RSK	RTH	X=FALSE Y=FALSE Z=FALSE
Left Ankle Power	LFT	LSK	X=FALSE Y=FALSE Z=FALSE	Right Ankle Power	RFT	RSK	X=FALSE Y=FALSE Z=FALSE
Left GRF	LFT	Virtual Lab	X=FALSE Y=FALSE Z=FALSE	Right GRF	RFT	Virtual Lab	X=TRUE Y=FALSE Z=FALSE

Oxford foot model

Joint	Segment/Reference	Negate?	Joint	Segment/Reference	Negate?
Left Hindfoot Tibia Angle	Left Hind-foot/Left Shank OFM	X=FALSE Y=TRUE Z=TRUE	Right Hind-foot Tibia Angle	Right Hind-foot/Left Shank OFM	X=FALSE Y=FALSE Z=FALSE
Left Forefoot Tibia Angle	Left Forefoot/Left Shank OFM	X=FALSE Y=TRUE Z=TRUE	Right Fore-foot Tibia Angle	Right Fore-foot/Left Shank OFM	X=FALSE Y=FALSE Z=FALSE
Left Forefoot Hindfoot Angle	Left Forefoot/Left Hindfoot	X=FALSE Y=TRUE Z=TRUE	Right Fore-foot Hindfoot Angle	Right Fore-foot/Left Hind-foot	X=FALSE Y=FALSE Z=FALSE
Left Hallux Forefoot Angle	Left Hallux/Left Forefoot	X=FALSE Y=FALSE Z=FALSE	Right Hallux Forefoot Angle	Right Hal-lux/Left Forefoot	X=FALSE Y=FALSE Z=FALSE

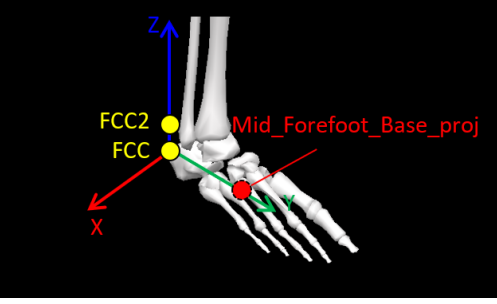
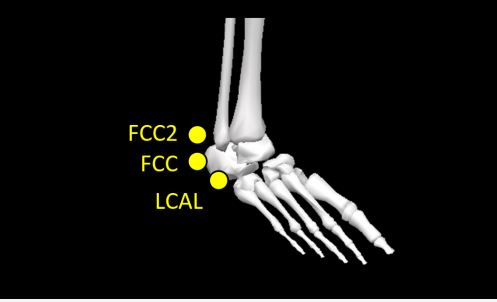
Appendix D - Oxford Foot Model

The Oxford Foot Model will be appended to the main model if the session setting "Multisegment Foot" is set to Oxford (see "Step 5 – Subject and Session details" on page 16). The material presented here is for the right foot, the left foot is defined accordingly.

Shank

To simplify the marker set, the original Oxford Foot Model definition of the shank is not used. Instead, the shank definition that is used for the lower body model is used, i.e. if a CAST marker set is used for the lower body, the CAST shank definition will be applied.

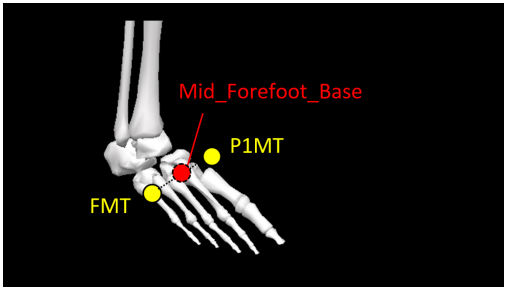
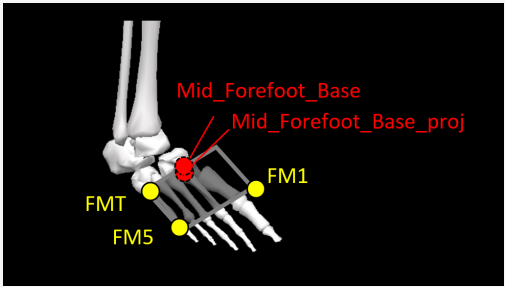
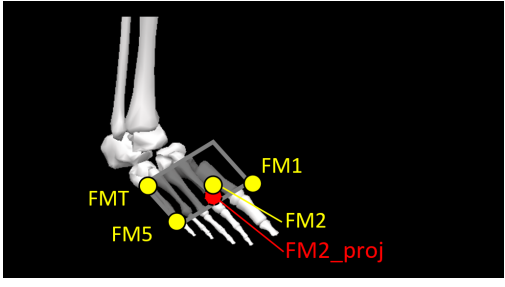
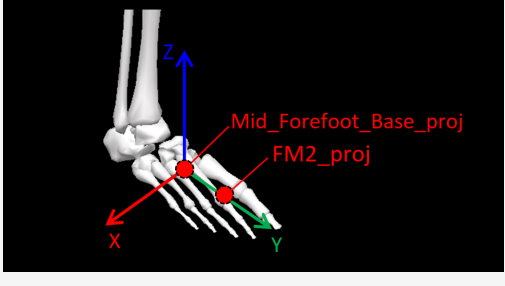
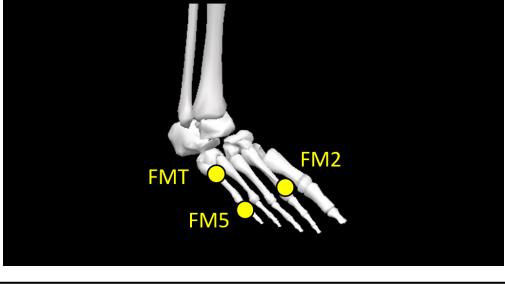
Hindfoot

Definition:	Figure:
The long axis of the foot (y-axis) is defined by the line connecting FCC and Mid_Forefoot_Base_proj (see Forefoot for definition of this landmark); the sagittal plane is defined by FCC2:	
Tracking markers:	

Reference

Stebbins et al. (2006), Variation 5 (p. 405). Note that the definition used by Qualisys has been adapted to use a projection of the Mid_Forefoot_Base landmark to define the long axis of the hindfoot, instead of assuming the axis to be horizontal. This was done so that the definition can be used for persons that cannot place their foot flat on the floor.

Forefoot

Definition:	Figure:
Mid_Forefoot_Base is midpoint of FMT and P1MT:	 A 3D model of a right foot skeleton. A red dot labeled 'Mid_Forefoot_Base' is positioned at the midpoint between a yellow dot labeled 'FMT' (first metatarsal base) and another yellow dot labeled 'P1MT' (proximal phalanx base of the first toe).
Mid_forefoot_base_proj is Mid_forefoot_base projected on plane defined by FMT, FM5 and FM1:	 A 3D model of a right foot skeleton. A red dot labeled 'Mid_Forefoot_Base' is shown. A red dot labeled 'Mid_Forefoot_Base_proj' is shown on a plane defined by three yellow dots: 'FMT', 'FM5', and 'FM1'. A line connects the base point to its projection.
FM2_proj is FM2 projected on a plane defined by FMT, FM1 and FM5:	 A 3D model of a right foot skeleton. A yellow dot labeled 'FM2' is shown. A red dot labeled 'FM2_proj' is shown on a plane defined by three yellow dots: 'FMT', 'FM1', and 'FM5'. A line connects FM2 to its projection.
The long axis of the forefoot (y-axis) runs from Mid_Forefoot_Base_Proj to FM2_proj; the orientation is defined so that the z-axis is perpendicular to the plane formed by FMT, FM1 and FM5:	 A 3D model of a right foot skeleton. A coordinate system is shown with a red arrow for the X-axis, a green arrow for the Y-axis, and a blue arrow for the Z-axis. The Y-axis runs from 'Mid_Forefoot_Base_proj' to 'FM2_proj'. The Z-axis is perpendicular to the plane formed by 'FMT', 'FM1', and 'FM5'.
Tracking markers:	 A 3D model of a right foot skeleton. Three yellow dots are shown, labeled 'FMT', 'FM5', and 'FM2', representing the tracking markers.

Reference

Stebbins et al. (2006), section 2.3.4 (p. 403).

Hallux

Definition:	Figure:
The landmark HLX_proj is at the x and y coordinates of FM1 and the z coordinate of PM6. This means that HLX_proj is at the same height relative to the floor as PM6 but below FM1:	
The Hallux_x landmark is a projection of HLX_proj in the X direction of the Forefoot coordinate system.	
The Y axis of the Hallux connects HLX_proj and PM6; the orientation is defined so that the transverse (X/Y) plane includes the Hallux_x landmark:	
Tracking markers: Note: D1M_HLX is a virtual marker. It is defined based on the static trial and at the position of FM1. It is tracked using the coordinate system of the forefoot segment.	

Reference

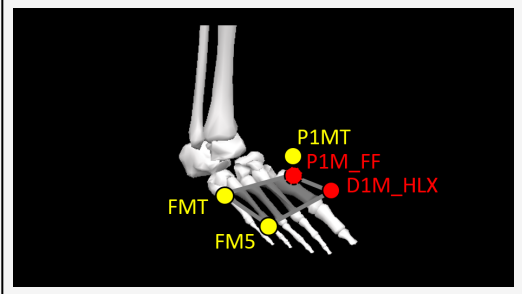
Stebbins et al. (2006), section 2.4.3 (p. 404)

Arch height

Arch height is calculated as the distance between P1MT and the landmark P1M_FF. The result is divided by the distance FM2 and FCC.

P1M_FF is a projection of P1MT onto the plane formed by FMT, FM5 and D1M_HLX.

Note: D1M_HLX is a virtual marker. It is defined based on the static trial and at the position of FM1. It is tracked using the coordinate system of the forefoot segment.



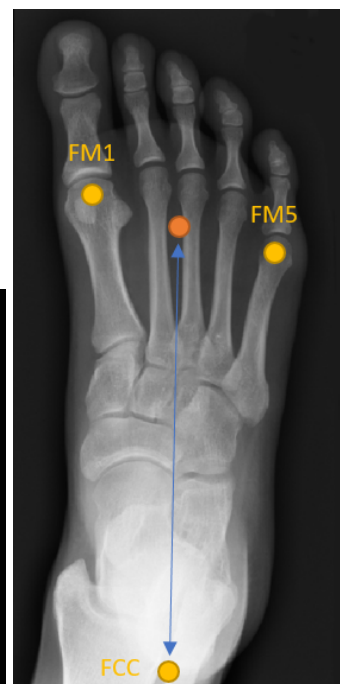
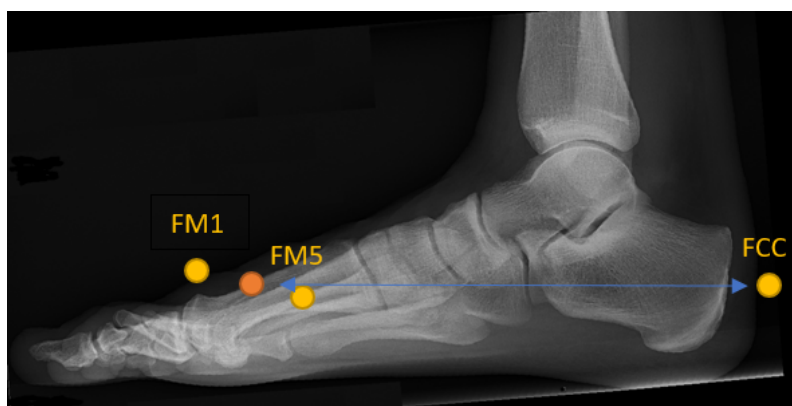
Reference

Stebbins et al. (2006), section 2.4.3 (p. 404)

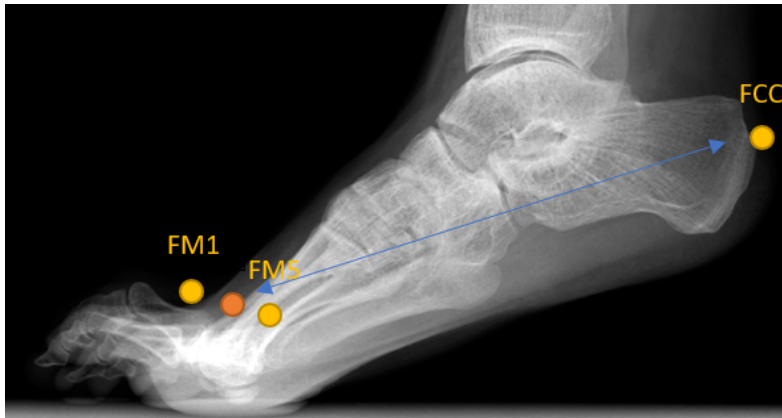
Foot maker placement compared to single segment foot

Single segment foot

- FM1/5 placed at dorsal aspect of metatarsal head (Capozzo 1995).
- FCC is placed at the same height as midpoint of FM1 and FM5. This is closer to “upper ridge of calcaneus”.

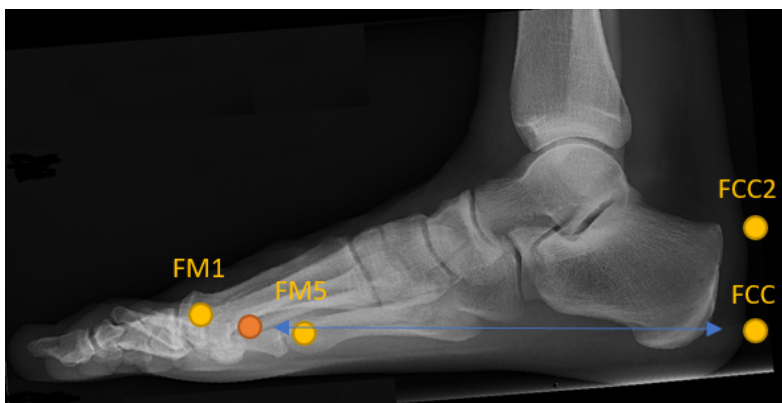


Equinus foot

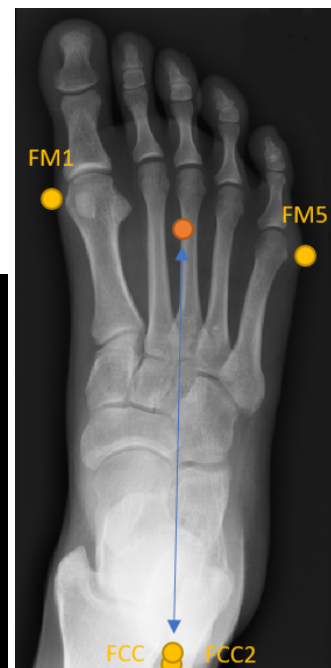


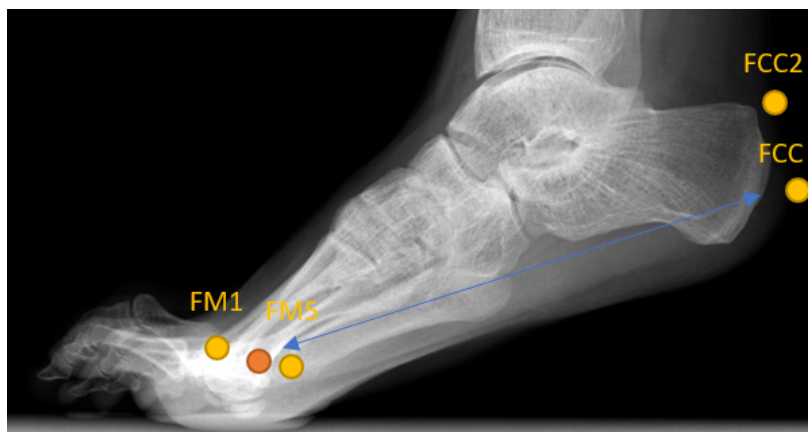
Oxford Foot Model

- FM1/5 are placed medially and laterally on the foot such that their centers lie along the line through the distal heads of the first and fifth metatarsal heads (Stebbins 2006).
- FCC is placed at the same height as midpoint of FM1 and FM5. FCC ends up lower and FCC2 can be placed at “upper ridge of calcaneus”.



Equinus foot

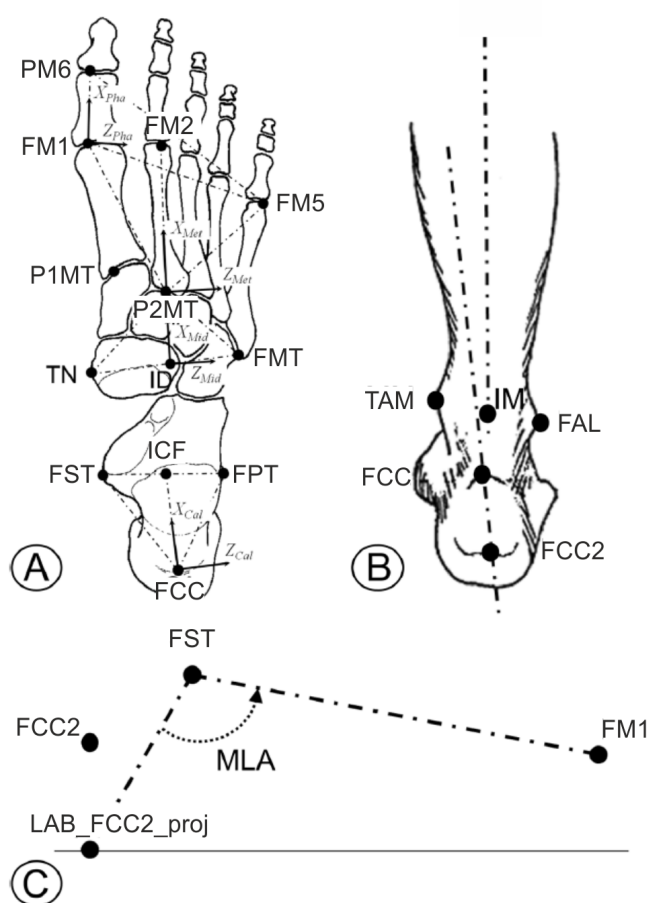




Appendix E - Rizzoli Foot Model

The Rizzoli Foot Model will be appended to the main model if the session setting "Multisegment Foot" is set to Rizzoli (see "Step 5 – Subject and Session details" on page 16).

For details of the model refer to Portinaro et al.: Modifying the Rizzoli foot model to improve the diagnosis of pes-planus: application to kinematics of feet in teenagers. Journal of Foot and Ankle Research 2014 7:57.



Adding synchronized video to Visual3D

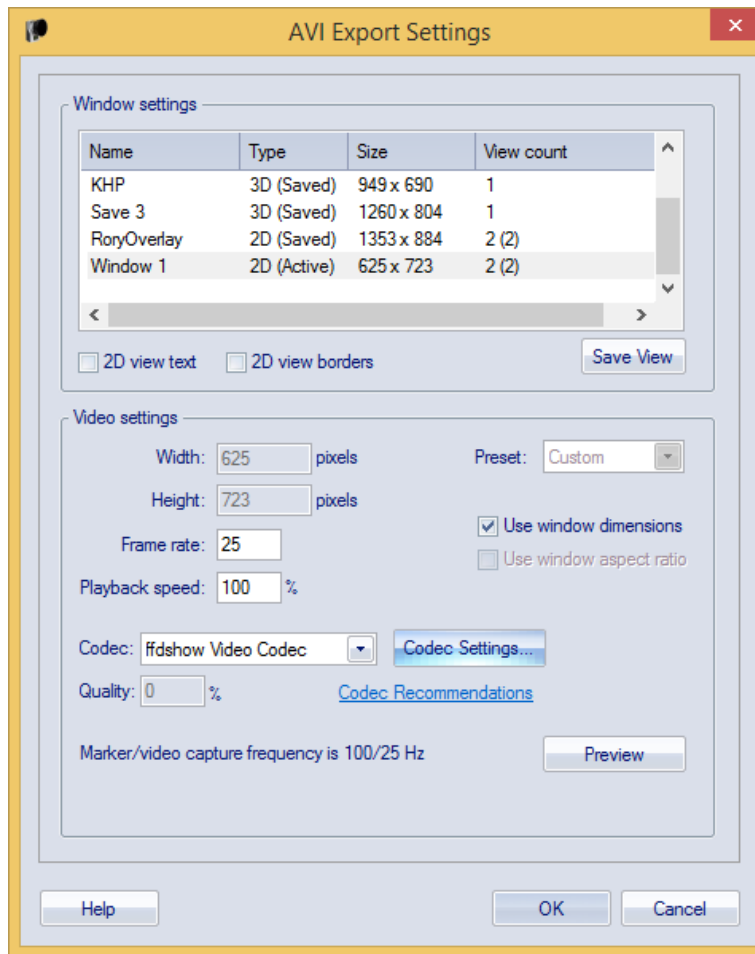
Often it is useful for view video files along with the animation and data in Visual3D. This tutorial outlines how this is done.

It is assumed that synchronized video was recorded together with dynamic trial. Video recorded by Oqus 2c is automatically synchronized with motion data.

1. In QTM open trial with the video included.
2. Switch to 2D view and select view(s) you want to include in final video (1). If you wish to include 3d overlay, right click on 2d camera view (2).



3. Navigate to File -> Export -> To AVI.
4. In “AVI Export Settings” select appropriate Window settings and Video settings.



Check the box **"Use window dimensions"**.

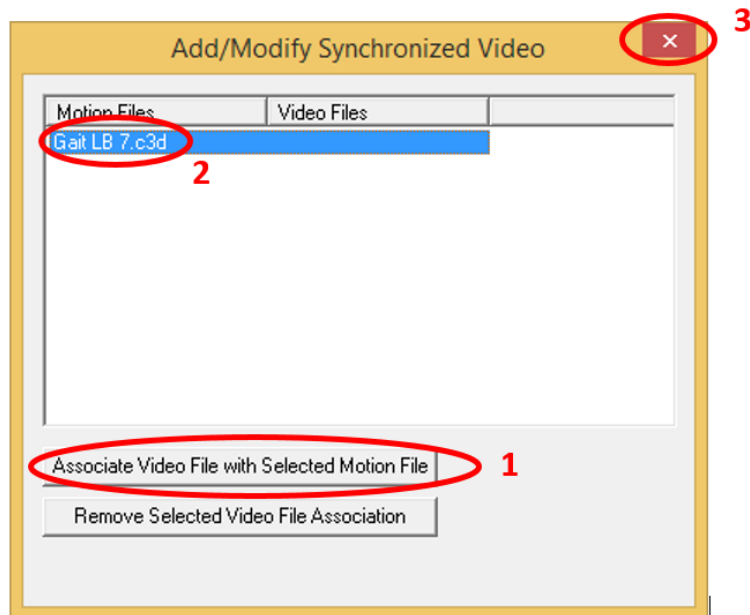
Frame rate: It is best to use the same frame rate that was used for recording the video.

Playback speed: Set to 100%.

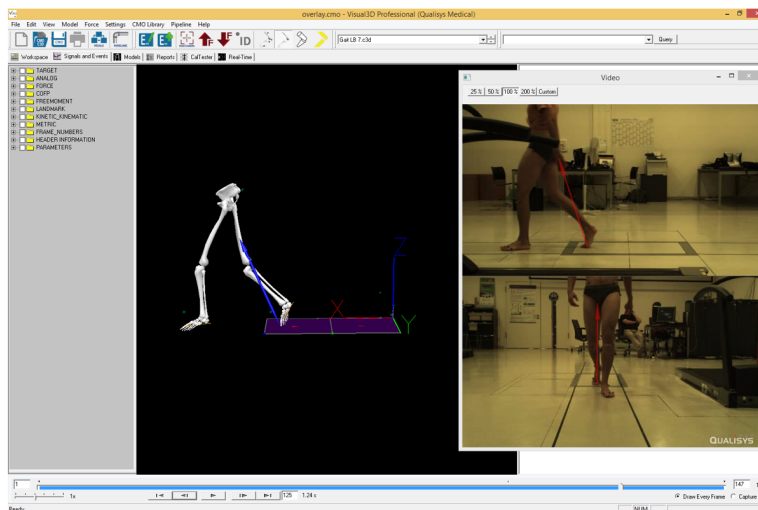
Codec: ffdshow is recommended, make sure it is set up correctly (see QTM help). Other Codecs may also be used but should be tested first.

Use QTM Help for further details on these settings.

5. In Visual3D, open cmo file and navigate to File > Add/Modify Synchronized Video Files.
6. Select correct motion files (1), click Associate Video File with Selected Motion File (2) and find the video exported from QTM. Close the window (3).



7. Make the file with associated video the active file. Video will appear. Now you can play video synchronously with the motion file. Use View > View synchronised video to turn the video window on or off.



Revision history

Gait 1.5.0

Changes since Gait 1.4.3

New features:

1. Added support for CGM2 marker set. CGM2 processing is done by QCGM2 engine. Visual3D is not required. See <https://pycgm2.github.io/> for details.
2. Reports with comparisons of multiple sessions can now be exported to PDF format in the online report platform.
3. Documentation and training material is now available in Portuguese.

Improvements:

1. Fields in module pane have been reorganized to better follow usual data handling workflow.
2. Forearm and hand model definition unified among models so that +X axis is pointing toward styloid process of radius (RSP) for the right arm and ulna (USP) for the left arm. For CAST and IOR only hand orientation was changed, for CGM both hand and forearm were changed.

Bug fixes:

1. Fix to Word report template to plot MAP header on new page.
2. Fix to default upper body reference data in offline web report.
3. Fix to annotations in online report to be able to place them at very bottom of graph.

Notes:

1. When updating from Visual3D v6 to Visual3D 2020, you must update the path to Visual3D in QTM > Project Options > Miscellaneous > Folder Options to C:\Program Files\Visual3D\Visual3D.exe

Known issues:

1. CGM2 processing has some limitations that will be addressed in future releases:
 1. Events have to be in correct sequence. Processing will fail if events for parts of the gait cycle are missing from the sequence.
 2. CGM2 processing works only for gait. Running is not supported.
 3. EMG data are not processed.

4. In case of gaps in timeseries signals, these gaps are set to zero instead of NaN (nodata).
5. 3d viewer is not shown.
2. Reference data shown in graphs is not from the same source as reference data used to compare temporal-spatial parameters.
3. Reference data for the web report cannot be customized so it will not change if the local normative file (.vnd) is changed.

Gait 1.4.3

Changes since Gait 1.4.0

New features:

Added support for Treadmetrix treadmill.

Improvements:

1. Normative sample was updated and new normatives were added for upper body and for Oxford Foot Model.
2. Model related changes:
 - Wrist is used as midpoint of ulna and radius.
 - Added option to use forearm marker (FAR).
 - Added option to use SXS and SNJ as tracking markers for thorax.
 - Oxford foot model - Hindfoot distal point defined from L/R_Mid_Forefoot_base_proj instead of L/R_FMT_proj. It makes definition more robust in case of foot deformities.
 - Convectional Gait Model - If R/L_FME and R/L_F_TAM exist, R/L_FME and R/L_TAM are used to define frontal plane of thigh and shank.
 - CAST, IOR - Normalized foot model changed to use projection of FCC, FM2 and FM5 to be consistent with Capozzo (1995).
 - CAST - Frontal plane based on maleoli only, and not maleoli and epicondyles. This is closer to Capozzo (1995) and avoids the frontal plane being somewhere between the knee and ankle axis.
 - Recommendation for placement of FCC and FCC2 marker was improved. See Appendix.

Bug fixes:

1. Bug fix causing that Foot Progression was not normalized to static when this option was selected in QTM pane field.

Known issues:

1. QTM 2019.3 supports Delsys SDK 3.5.x (currently 3.5.3) but it is required to check the "backwards compatibility" option under Configure > Digital output. Otherwise, there will be synchronization errors.
2. Reference data shown in graphs is not from the same source as reference data used to compare temporal-spatial parameters.

Gait 1.4.2

Changes since Gait 1.4.1

New features:

None.

Improvements:

None.

Bug fixes:

1. Fix to issue where online web report could not be generated (PHP error message appeared).

Known issues:

1. Same as in previous release.

Gait 1.4.1

Changes since Gait 1.4.0

New features:

None.

Improvements:

1. Conventional gait model: If FME and TAM markers exists, they are used to define frontal plane of thigh and shank.

Bug fixes:

1. Fix to issue with loading EMG signals if EMG data are collected in separate trials.
2. Fix to bug preventing the use of custom EMG normatives.

Known issues:

1. Same as in previous release.

Gait 1.4

Changes since Gait 1.3.2

New features:

1. New online version of web report. Process session in Visual3D ("Visual3D Analysis"), then select "Online web report" to upload the data. Use your qualisys.com credentials to log in to the Report Center to retrieve reports (<https://report.qualisys.com>).
2. EMG data from Myon/Cometa devices can be processed and displayed in the reports.
3. Option to add the Rizzoli Foot Model to any of the body marker sets (IOR, CAST, CGM).

Improvements:

1. Online gait report has been adapted to be viewable on mobile devices.
2. Online gait report has a new feature to highlight a gait trial ("representative trial").
3. Online gait report displays the first video and 3D viewer when opening a file. The next time the report is loaded, the browser remembers last choice.
4. Oxford Foot Model: The landmark that is used to define the distal end of the calcaneus segment has been changed to coincide with the proximal landmark of the forefoot. Background: with the previous definition it was possible that the P1MT marker was projected to a position below floor level in cases where the person could not stand with neutral ankle varus/valgus in the static trial, resulting in the long axis of the calcaneus having a steep slope).

Bug fixes:

1. Improved handling of special EMG setups in the online web report (e.g. EMG on one body side only).
2. Removed TTC marker from the CGM marker guide as it was not included in AIM model.
3. Static trial for Left Hip Flexion was plotted in bold (Visual3D report).
4. Removed inconsistencies regarding Forearm marker placement between marker guide, AIM and V3D model.
5. Fixed an error in MAP calculation for comparison report.
6. Fixed an issue where the calculated step length was incorrect when using instrumented treadmill.
7. Fixed an issue where timeseries.xml was not generated in specific cases, resulting in an error during report generation.
8. Fixed incorrect spelling of IQR in Visual3D report (IRQ instead of IQR).
9. "ACC" signals in online web report are no longer displayed if Delsys EMG is used.

10. Fixed playback speed of 0.05 in online web report (the slowest speed is 1/16 now).
11. Corrected total count of markers in CAST marker guide.

Known issues:

1. 3D viewer in web report shows bones in wrong order if any body segment is missing throughout the entire trial. This is a visualisation issue only and does not affect the calculated parameters in the report.
2. 3D viewer in web report displays a maximum of two force plates. This is a visualisation issue only and does not affect the calculated parameters in the report.
3. QTM does not check if session file (.r3w, formerly .cmo) was edited by user and it is overwritten without warning.
4. When single belt treadmill is used, moment and power graphs in web report show entire signal from heel strike to heel strike. This makes moments and powers inaccurate between heel strike and opposite toe off due to double limb support.

Gait 1.3.2

Changes since Gait 1.3.1

New features:

None.

Improvements:

None.

Bug fixes:

1. Removed unnecessary folder (named "Installer") from project.

Known issues:

1. 3D viewer in web report shows bones in wrong order if any body segment is missing throughout the entire trial.
2. QTM does not check if session file (.r3w, formerly .cmo) was edited by user and it is overwritten without warning.
3. When single belt treadmill is used, moment and power graphs in web report show entire signal from heel strike to heel strike. This makes moments and powers inaccurate between heel strike and opposite toe off due to double limb support.

Gait 1.3.1

Changes since Gait 1.3.0

New features:

None.

Improvements:

1. Internal restructuring to accommodate specific customizations.

Bug fixes:

None.

Known issues:

1. 3D viewer in web report shows bones in wrong order if any body segment is missing throughout the entire trial.
2. QTM does not check if session file (.r3w, formerly .cmo) was edited by user and it is overwritten without warning.
3. When single belt treadmill is used, moment and power graphs in web report show entire signal from heel strike to heel strike. This makes moments and powers inaccurate between heel strike and opposite toe off due to double limb support.

Gait 1.3.0

Changes since Gait 1.2

New features:

1. Possibility to use Reckon3D as an biomechanical engine instead of Visual3D.
2. Several new features in web report (playheads in graphs, adding features dialog, supplementary data, more informative tooltips when hovering over items, clinical examination page).
3. Web report and user documentation are now multi-lingual (Swedish, German, French, Spanish).

Improvements:

1. Possibility to use more general naming for EMG signals in web report.
2. Added events to multisegment foot graphs in Visual3D report.
3. Improvements to video encoding if cropping files.
4. Showing numerical values when hovering over the curve in web report.

5. Added frame-by-frame scrolling through the videos and better 3D viewer controls.
6. Showing static trial in consistency graphs in web report.
7. Highlighting the current cycle curing playback in web report.
8. Showing hand graphics in 3D viewer of web report if hand makers exist.
9. Force vector is shown in 3D viewer of web report.
10. Several other small improvements.

Bug fixes:

1. Hide elbow rotation graphs in Word report for CGM model.
2. Fix the small differences in temporal-spatial parameters between Word and web report.
3. Fix to missing RTO events in some graphs in Word report.
4. Many other small fixes.

Known issues:

1. 3D viewer in web report shows bones in wrong order if any body segment is missing throughout the entire trial.
2. QTM does not check if session file (.r3w, formerly .cmo) was edited by user and it is overwritten without warning.